Azure Sentinel Brute-Force Detection Lab

This project shows how to use Microsoft Azure and Microsoft Sentinel to construct a cloud-based Security Information and Event Management (SIEM) solution from start to finish. The lab's objective was to use security event ingestion, KQL queries, and a custom analytics rule to successfully detect a brute-force attack on a Windows virtual machine.

- **This Lab Covers:**
    - Setting up Azure infrastructure
    - Installing a VM for Windows
    - Setting up the Log Analytics Workspace
    - Azure Monitor Agent (AMA) installation
    - Collecting Security Events for Windows
    - Simulating brute-force login activities
    - Creating detection-focused KQL searches
    - Creating a scheduled analytics rule
    - Generating and investigating into security events
- **This Projects Demonstrates Practical SOC-Level Skills Including:**
    - Ingestion and normalisation of logs
    - Engineering for threat detection
    - Writing KQL queries
    - Generating and investigating security incidents
    - Architecture for security monitoring

The SIEM pipeline was correctly configured and operating as anticipated, as evidenced by the successful creation of an incident alert for several unsuccessful login attempts.

Threat Context: Why Brute Force Matters

In on-premises and cloud environments, brute-force attacks are a popular credential access method. In order to obtain initial access, threat actors commonly target internet-exposed services like RDP and make many authentication attempts.

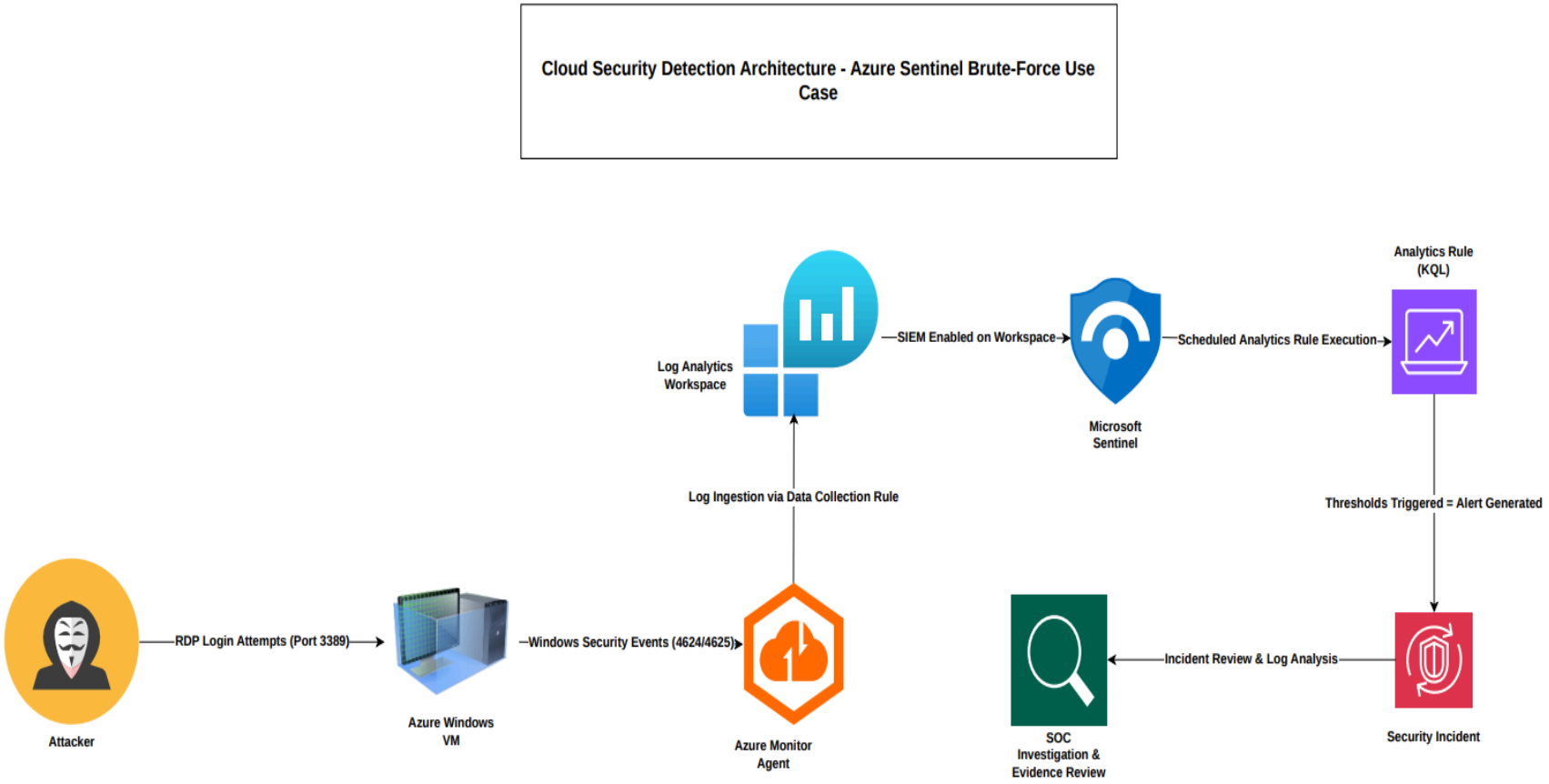
Multiple failed attempts at logging in may be a sign of automated brute-force activity, credential stuffing, or password spraying in enterprise settings. It may be difficult to identify this activity in the absence of centralised monitoring and detection methods.

Using Windows Security Event logs (Event ID 4625) to detect unusual authentication activity and transforming that information into actionable security incidents through Microsoft Sentinel are the main objectives of this lab.

MITRE ATT&CK Mapping

- **Tactic:** Credential Access
- **Technique:** Brute Force (T1110)
- **Sub-technique:** Password Guessing (T1110.001)

Cloud Security Detection Architecture - Azure Sentinel Brute-Force Use Cases



Phase 1: Creating the Resource Group

A dedicated Azure Resource Group was established based on figures 1-3 in order to logically contain and oversee all of the lab's infrastructure components, ensuring a well-organised deployment and streamlined resource management.

Figure 1. Microsoft Azure Homepage

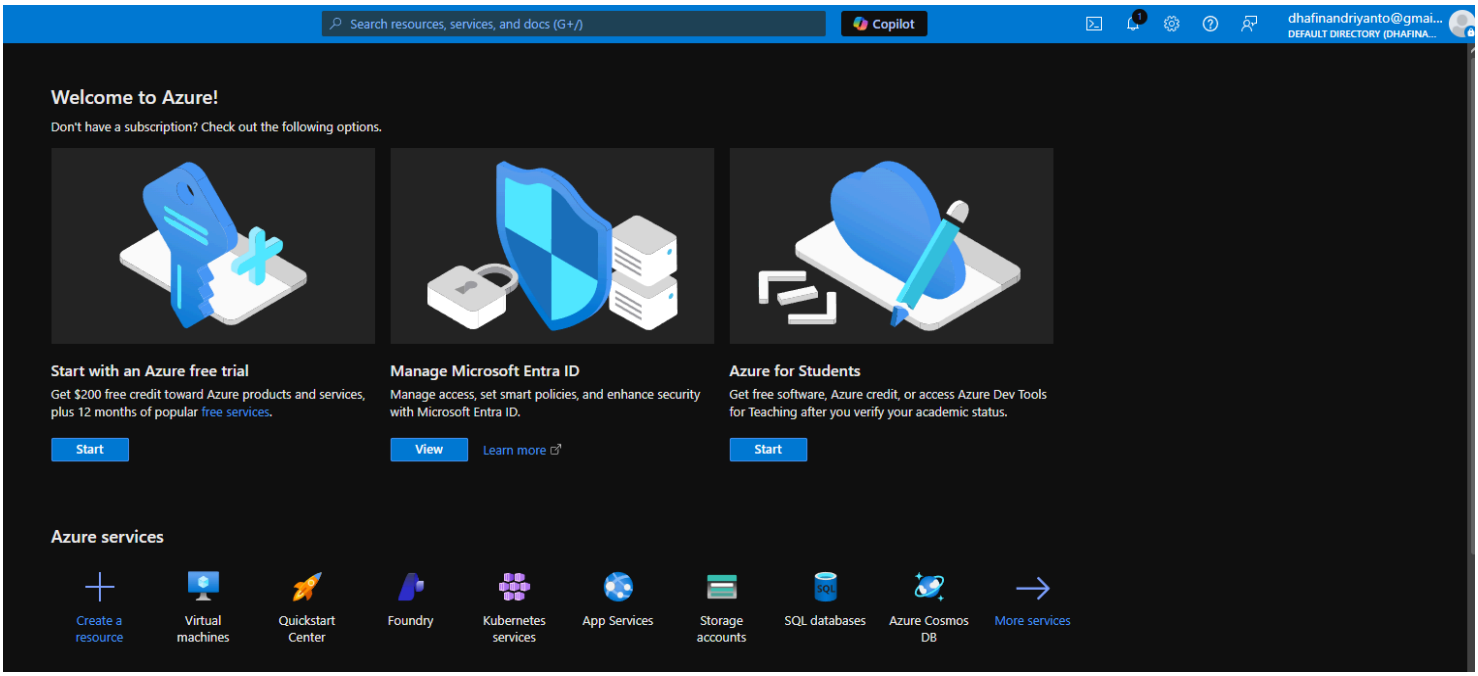


Figure 2. Creating a Resource Group for Azure Sentinel

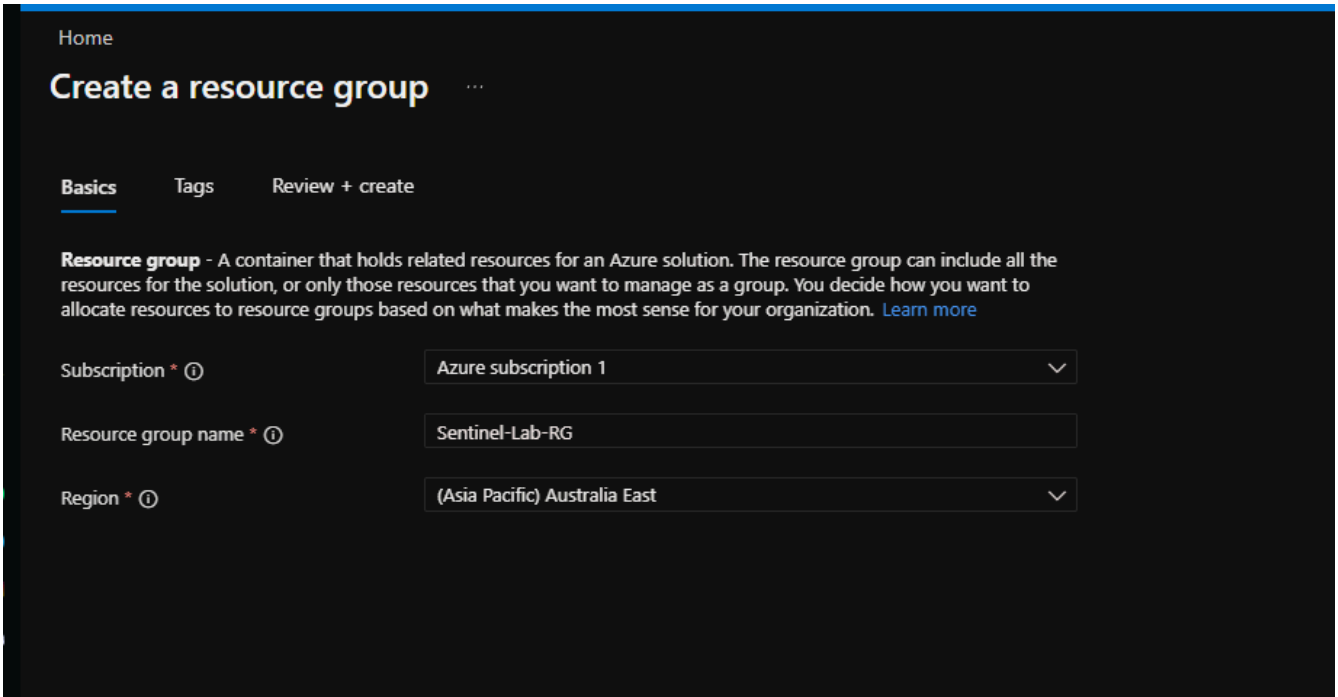
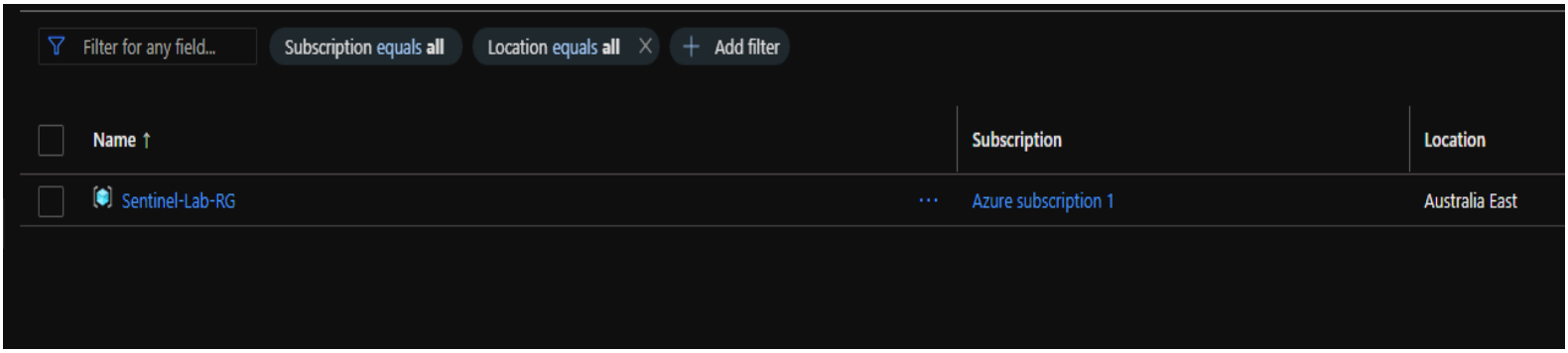


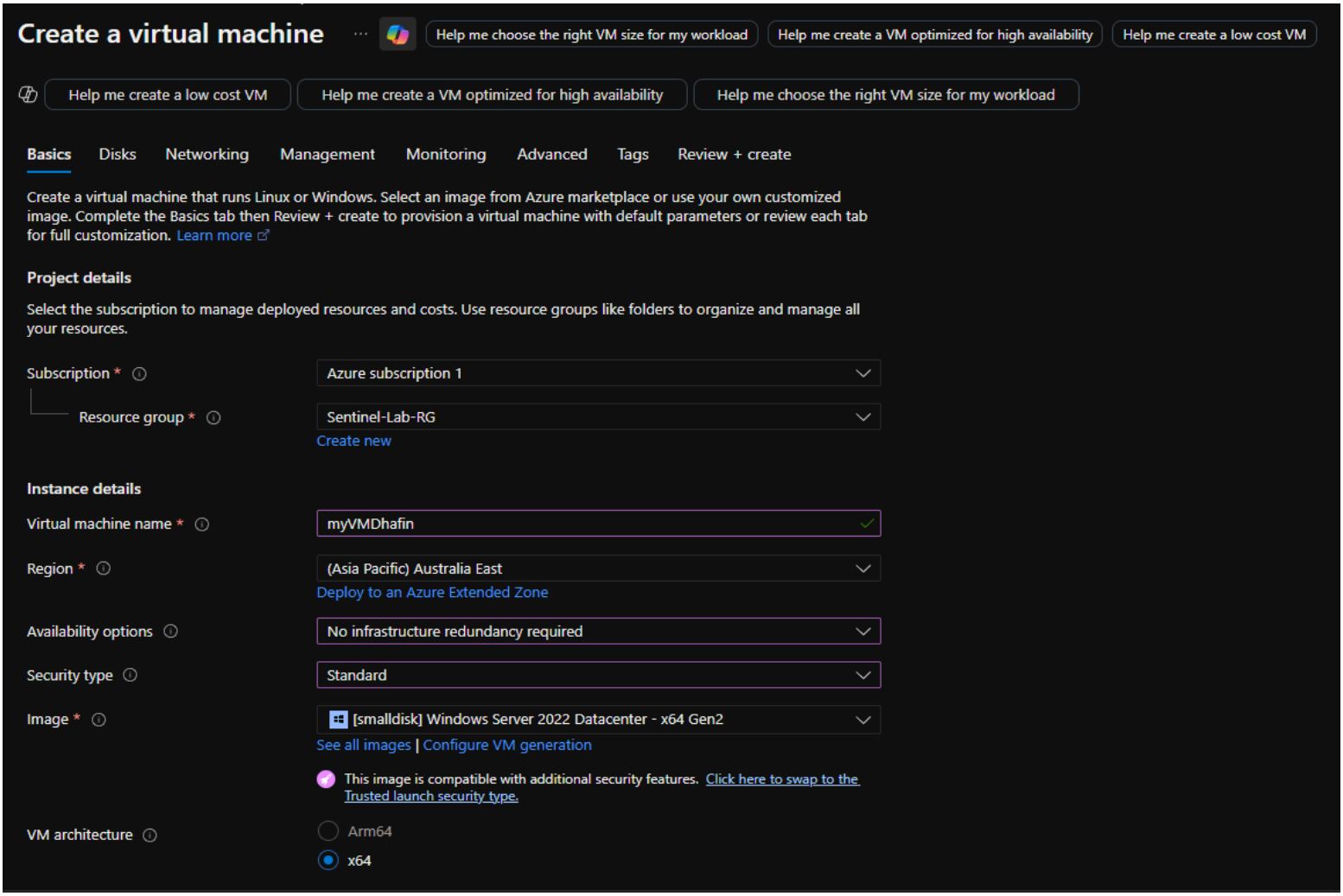
Figure 3. Successful Deployment of Resource Group



Phase 2: Creating the Windows Virtual Machine

Within the resource group, a Windows Server virtual machine was set up as the target endpoint for brute force attack simulation and authentication monitoring.

Figure 1. Filling in Details for the Virtual Machine Creation



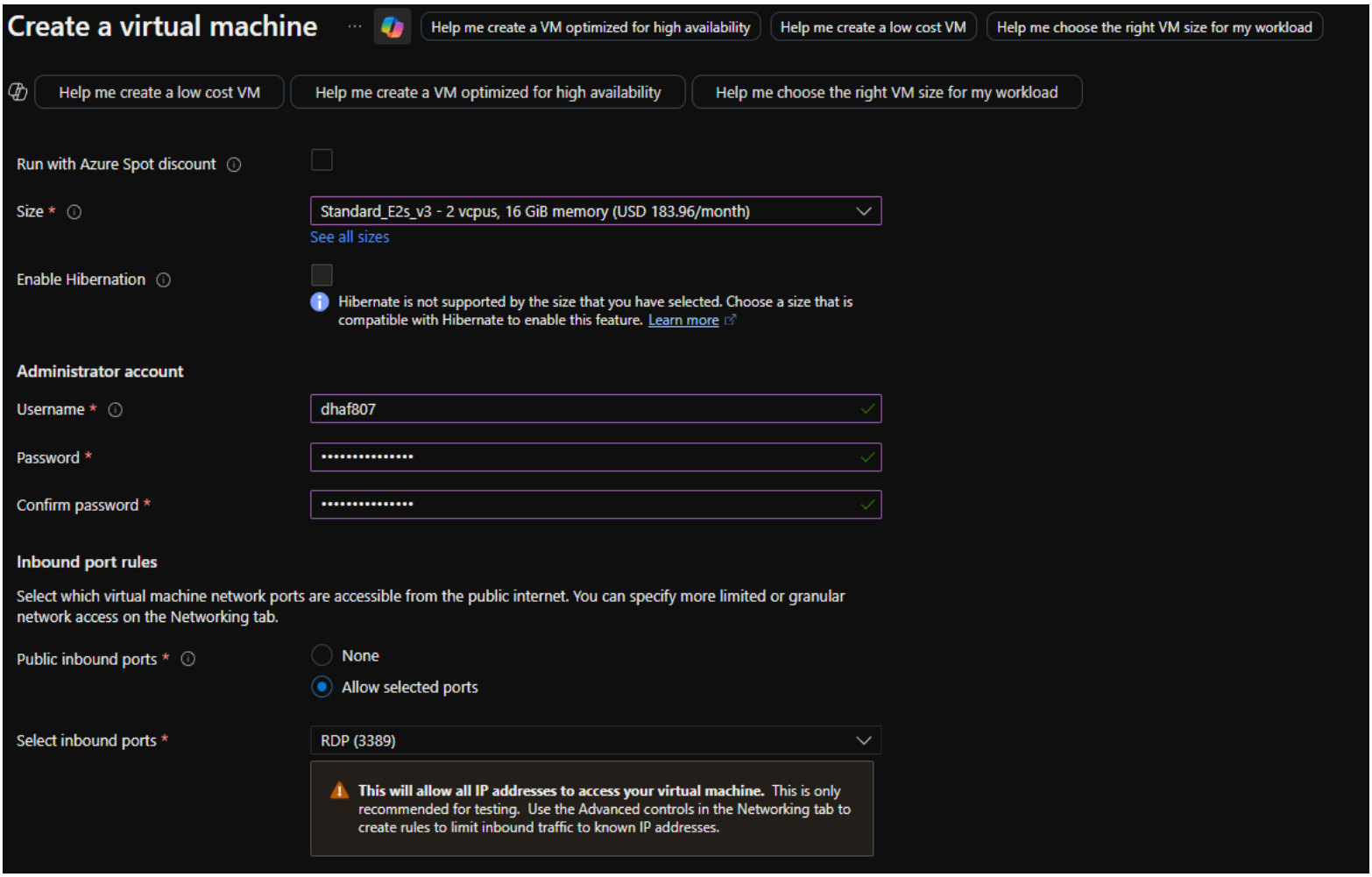


Figure 2. Completion of Deployment (Windows VM)

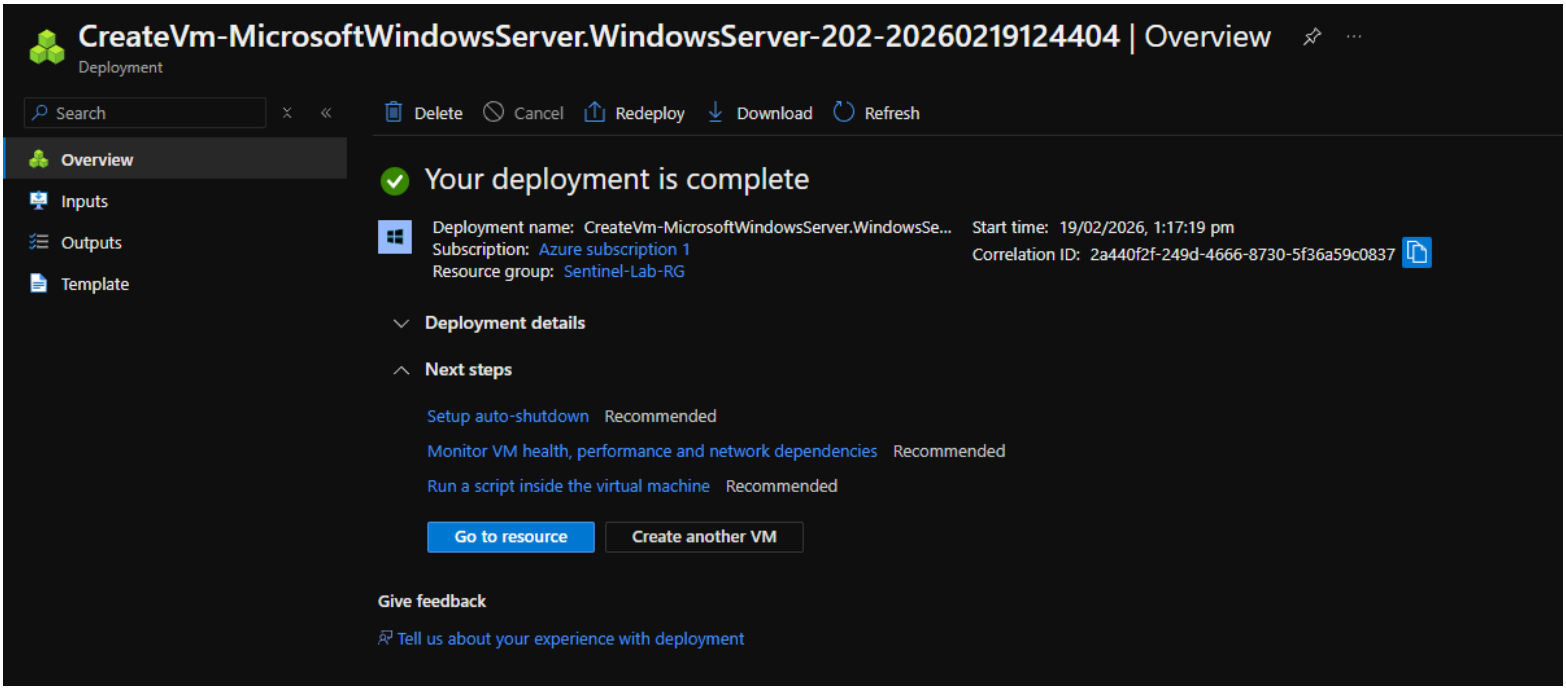


Figure 3. VM in Resource Group

Sentinel-Lab-RG
Resource group

Export resource groups using Bicep or Terraform | Help me generate Terraform for this resource group configuration. | +1

Search

Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Monitoring
Automation
Help

Create | Manage view | Delete resource group | Refresh | Export to CSV | Open query | ...

Essentials

Subscription (move) : Azure subscription 1
Subscription ID : 3036a6d5-0bfd-427a-b7fa-ee48ef2438d5
Tags (edit) : Add tags
Deployments : No deployments
Location : Australia East

Resources | Recommendations

Filter for any field... | Type equals all | Location equals all | Add filter

Name ↑	Type	Location
myVMDhafin	Virtual machine	Australia East
myVMDhafin-ip	Public IP address	Australia East
myVMDhafin-nsg	Network security group	Australia East
myvmdhafin552	Network Interface	Australia East
myVMDhafin_OsDisk_1_59ef9b12ac7e40ce8068643fb429fbd2	Disk	Australia East
vnet-australiaeast	Virtual network	Australia East

Phase 3: Testing and Connecting to VM

RDP was implemented to access the virtual machine to confirm its functionality and successful deployment. The environment was configured to enable testing of the authentication events needed to validate detection.

Figure 1. Connecting to VM and Downloading RDP File:

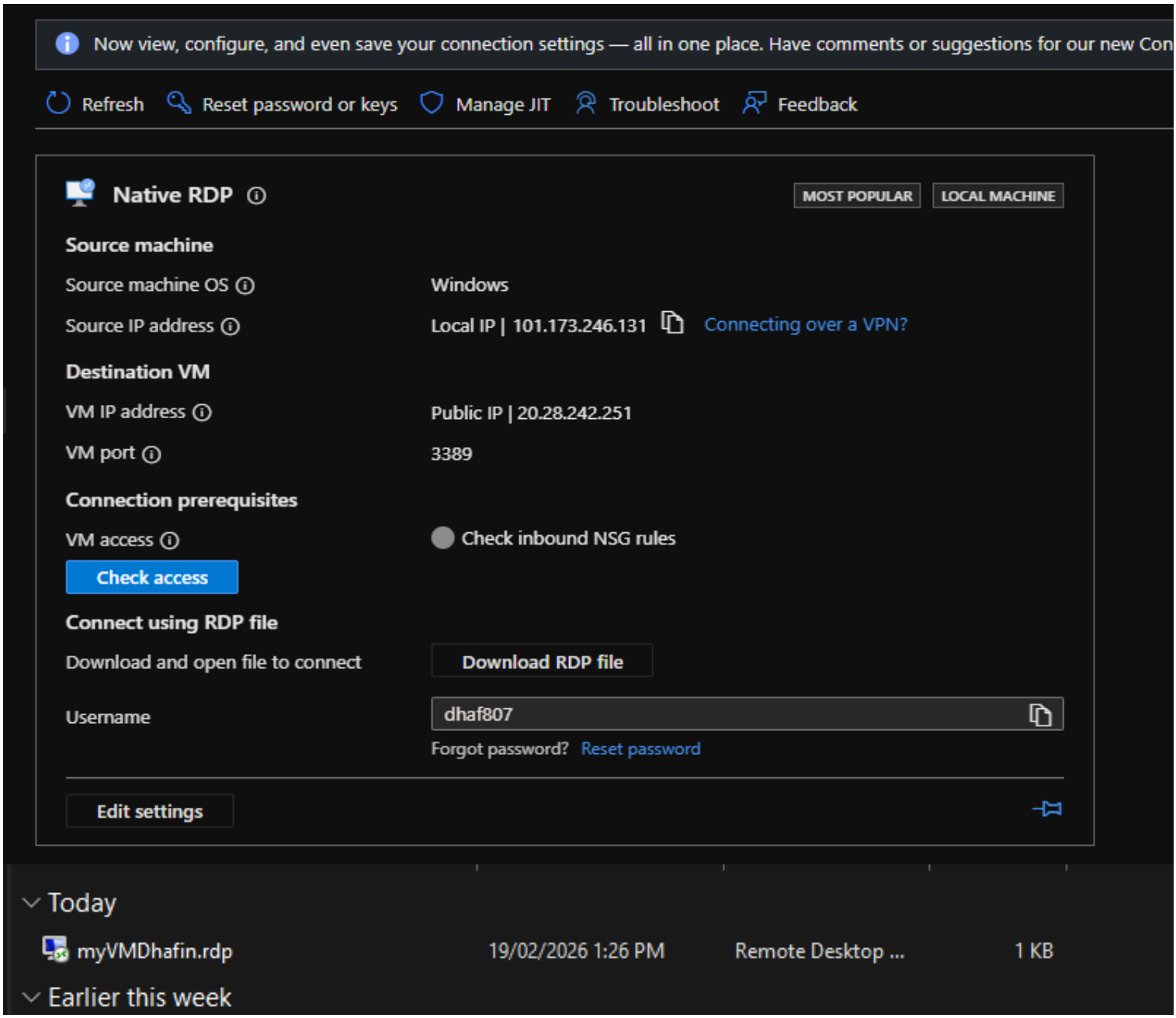


Figure 2. Entering Credentials for Connection and Access

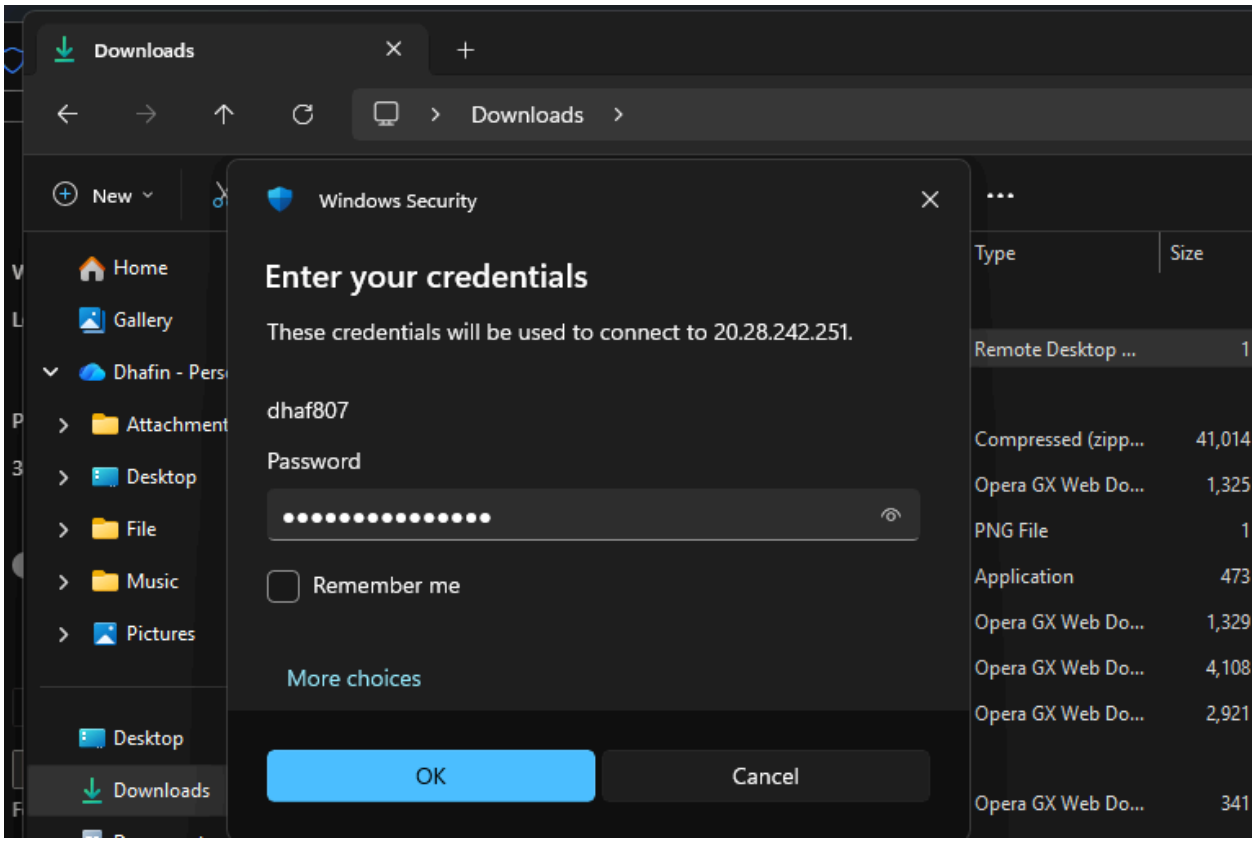


Figure 3. VM Functioning Properly After Entering Credentials (Success)

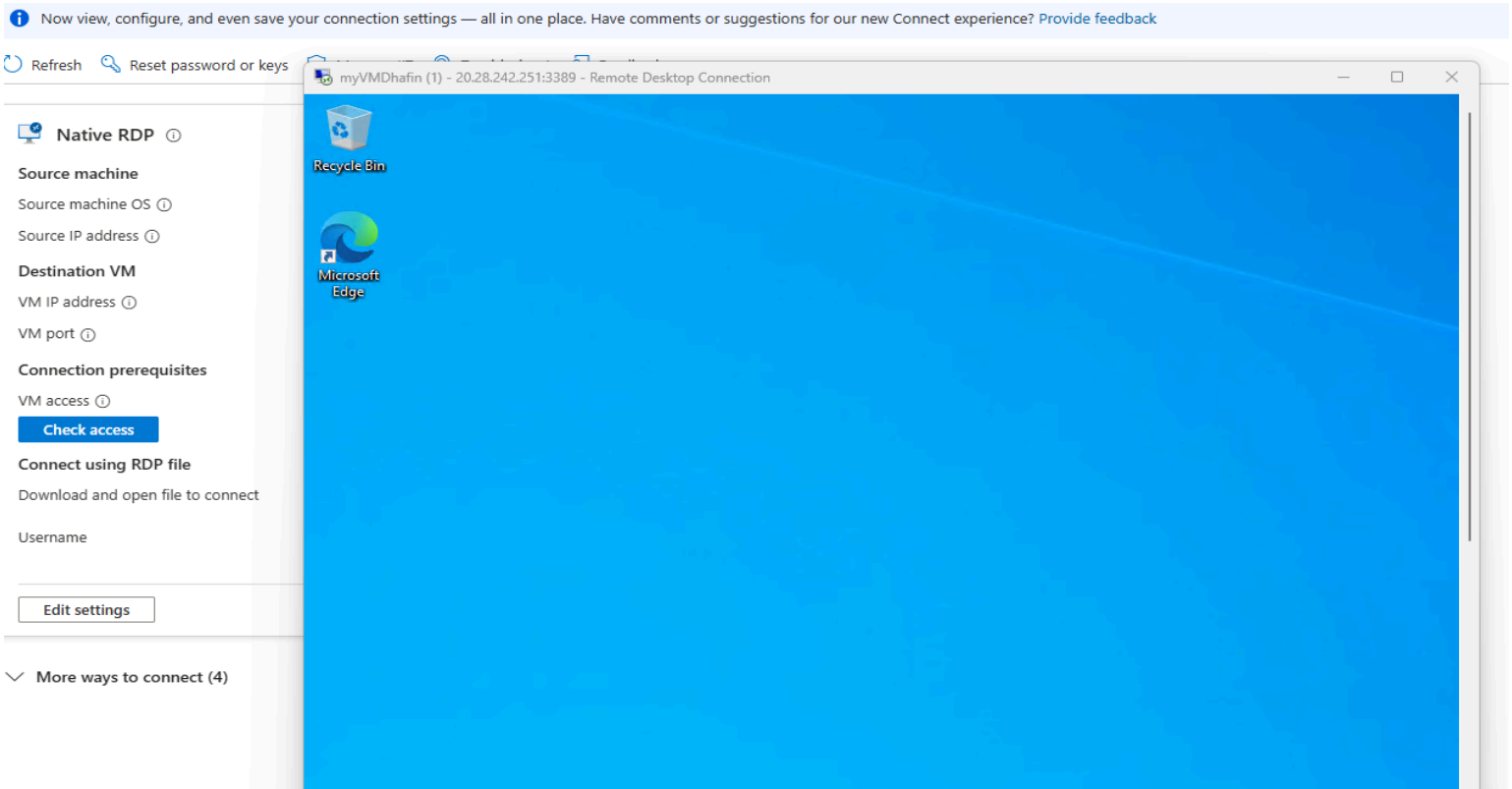
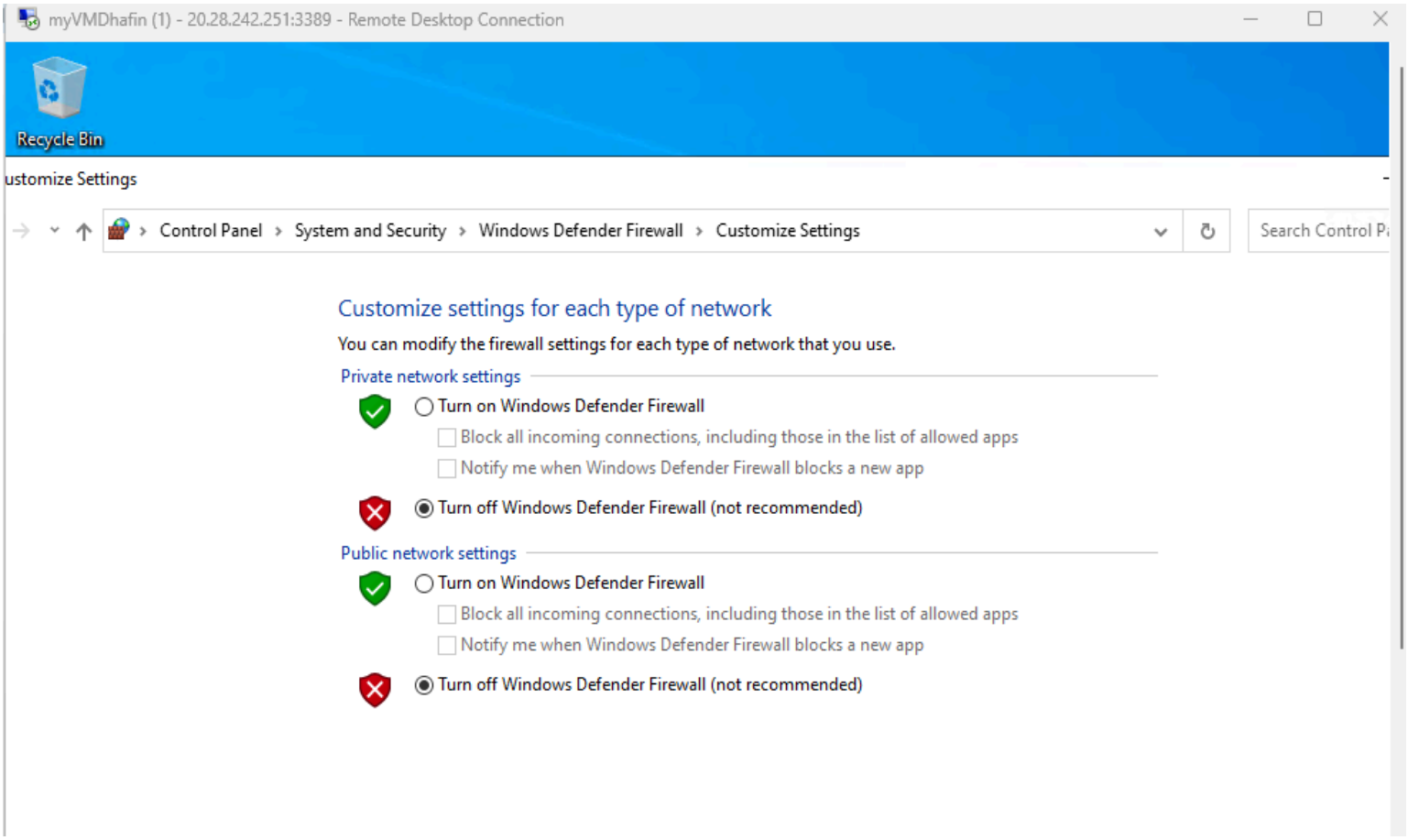


Figure 4. Turning off Windows Defender Firewall for Testing



Phase 4: Creating a Log Analytics Workspace

To centralise the ingestion, storage, and querying logs produced by the virtual machine, a Log Analytics Workspace was deployed.

Figure 1. Filling in Details for Log Analytics Workspace

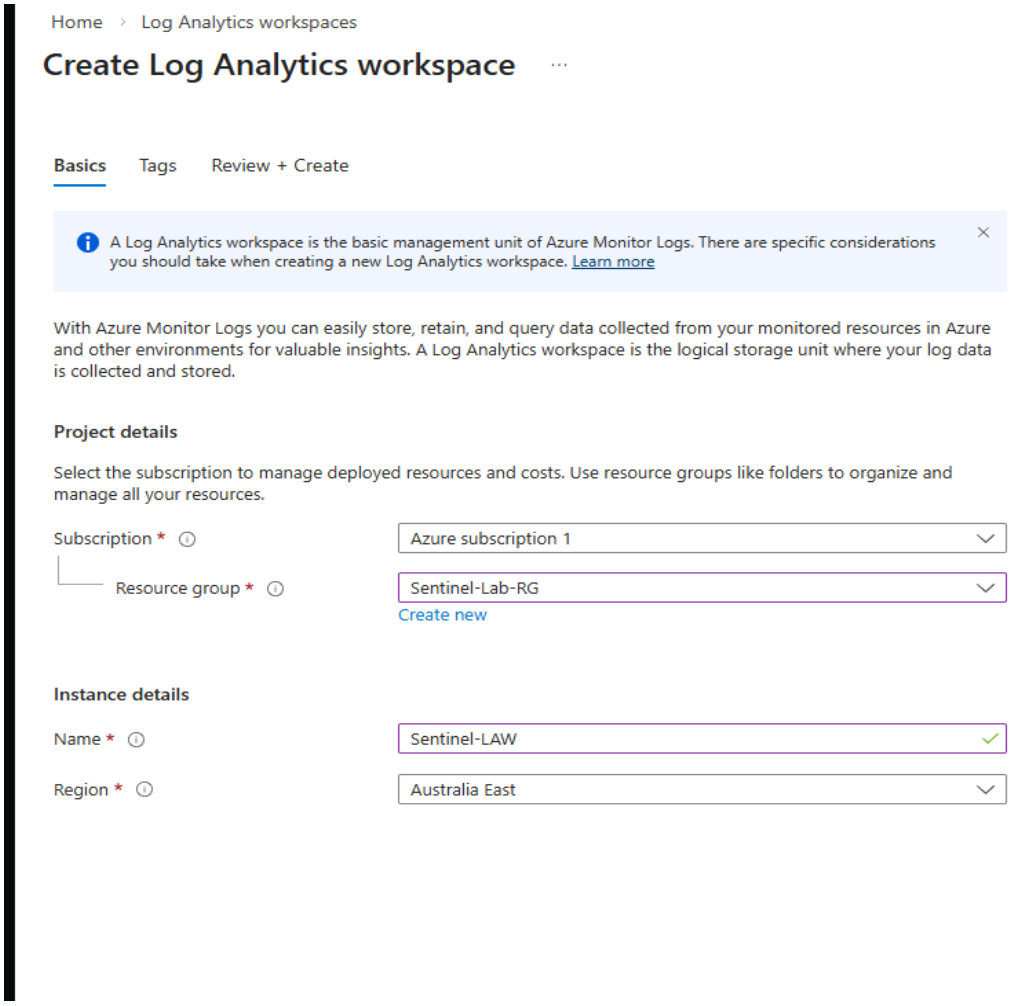
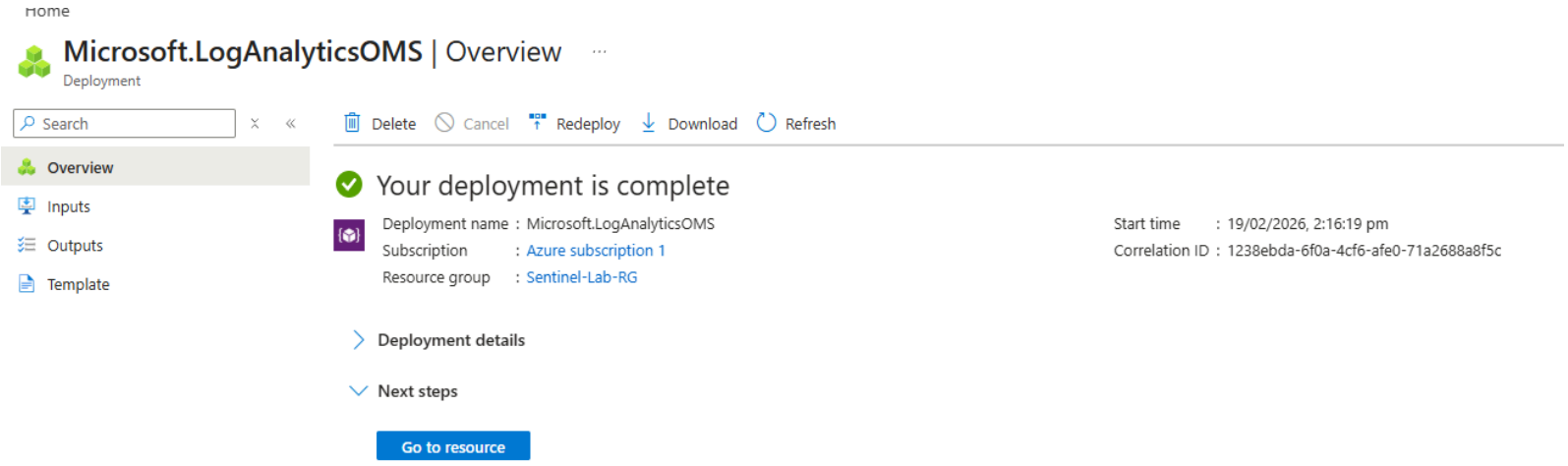


Figure 2. Log Analysis Workspace Deployment Completed



Phase 5: Installing Azure Monitoring Agent onto VM

For monitoring and analysis, a Data Collection Rule was configured to ingest Windows Security Events into the Log Analytics Workspace when the Azure Monitor Agent was installed

Figure 1. Filling in the Details for the Data Collection Rules

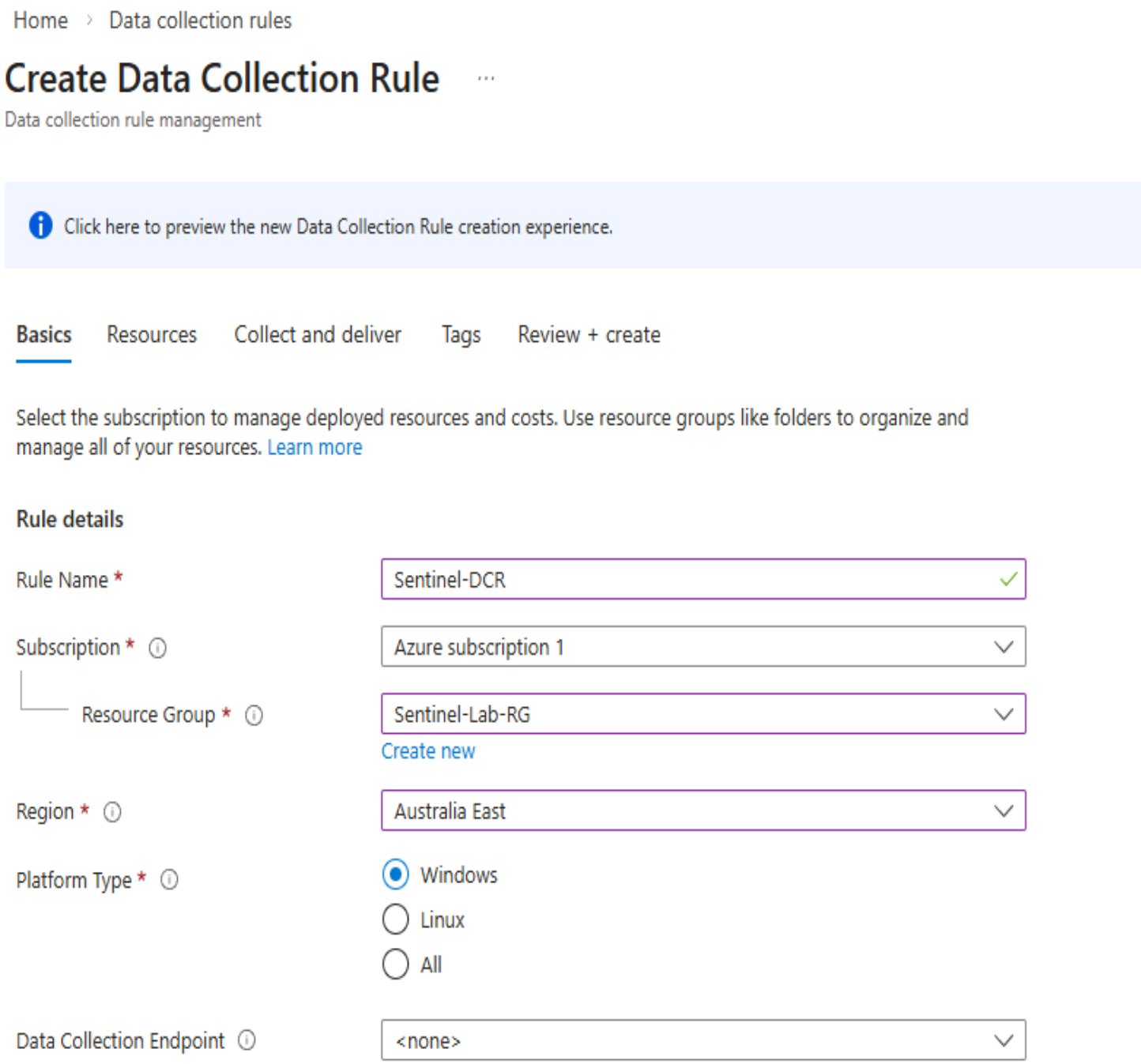


Figure 2. Adding the Windows VM as a Resource

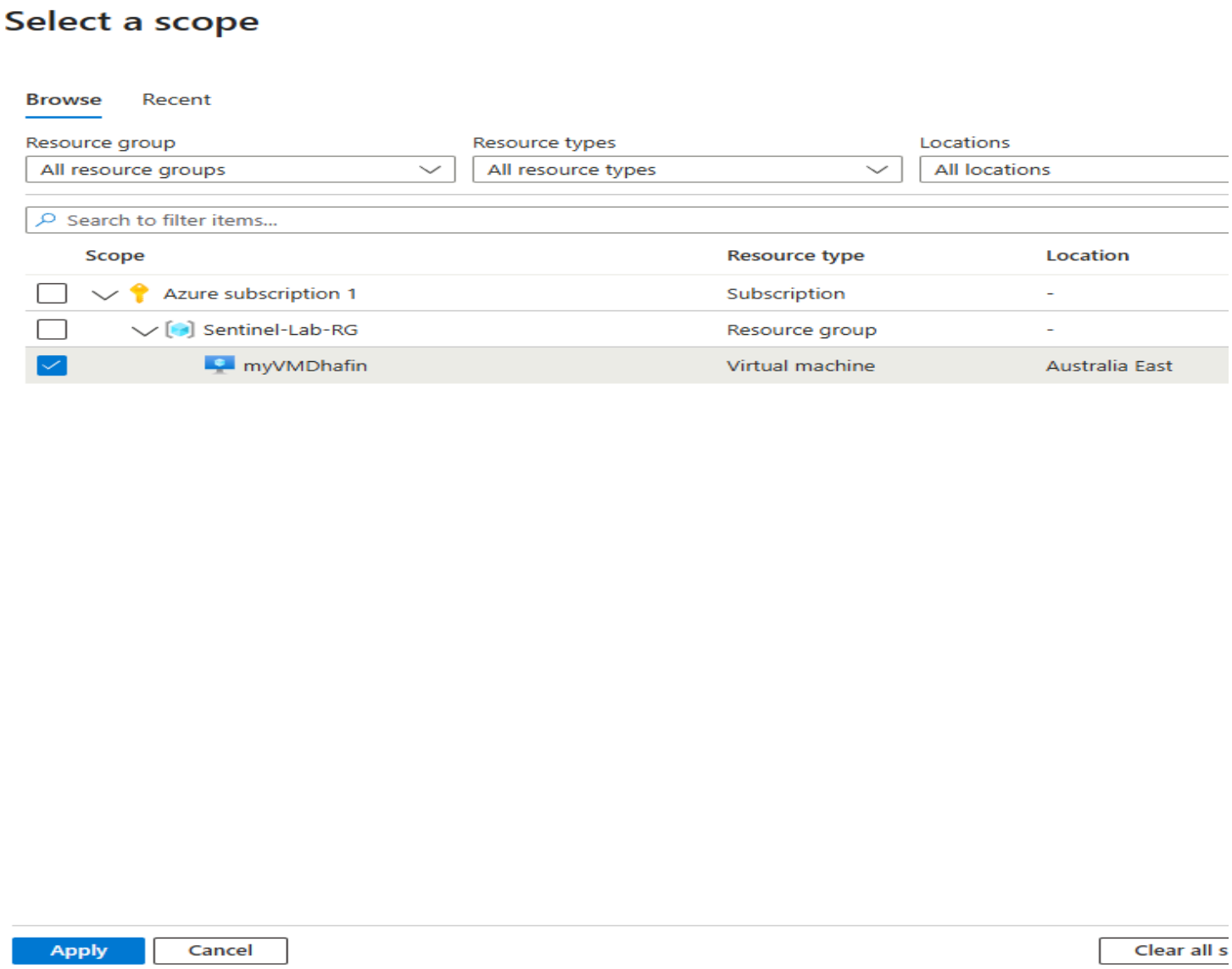
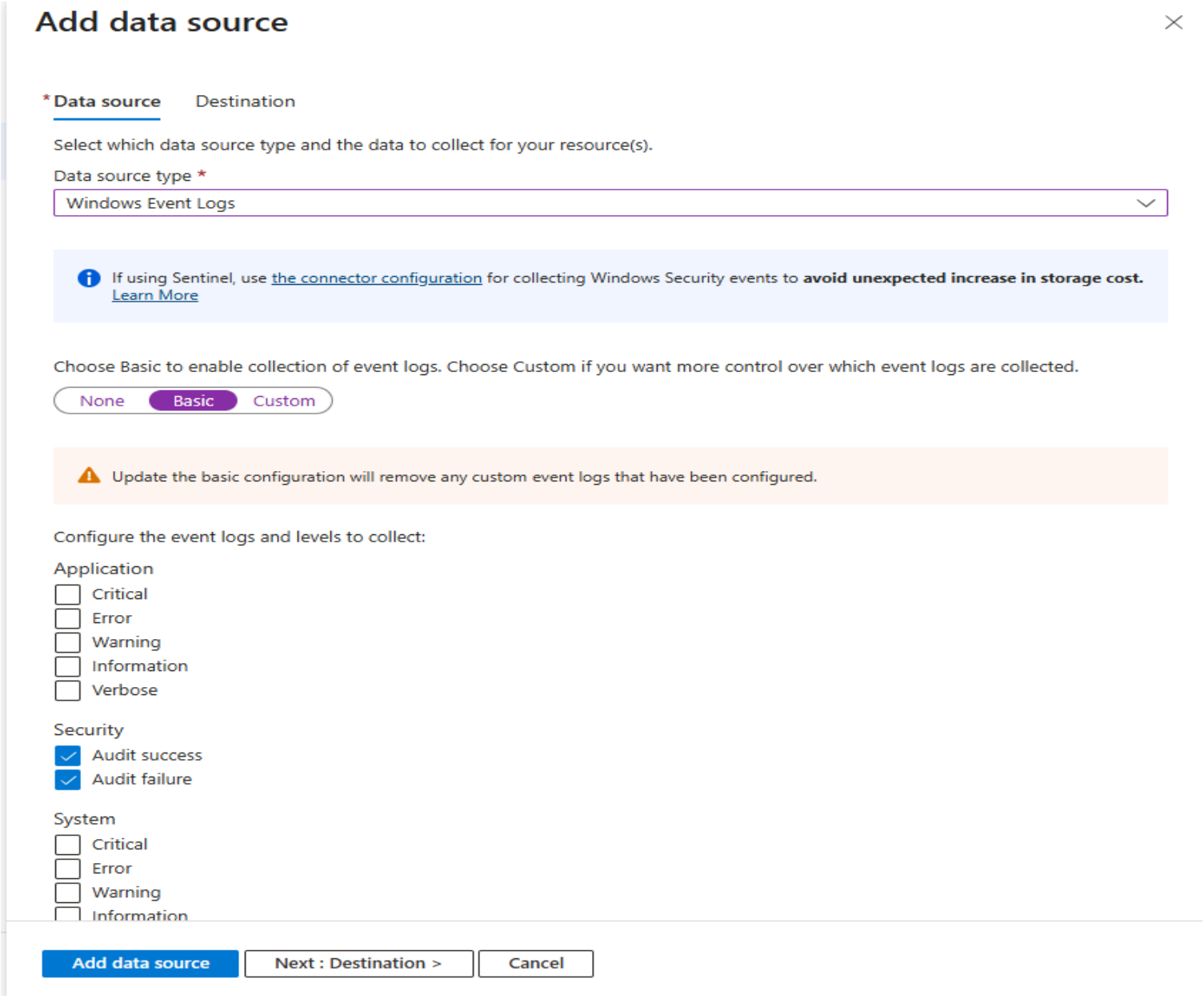


Figure 3. Adding a Data Source



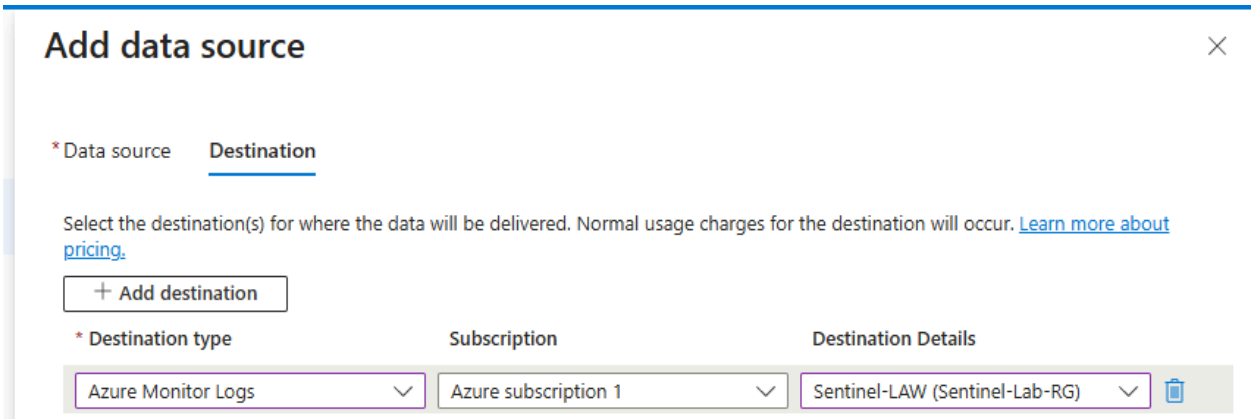


Figure 4. Data Collection Rules Deployment Completed

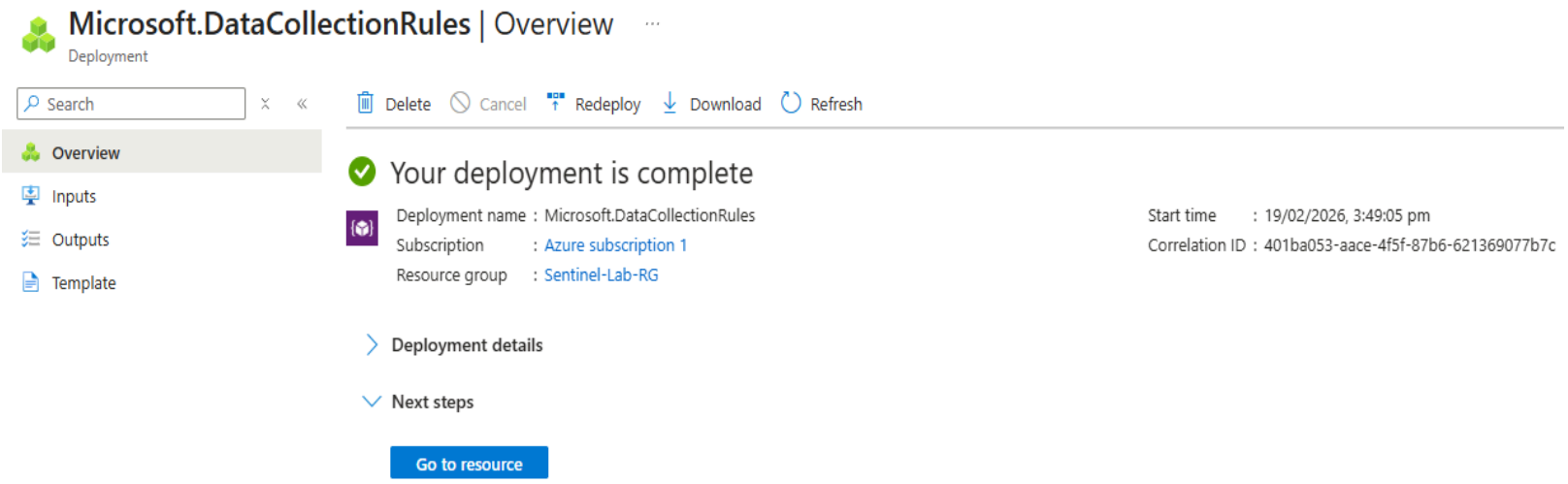


Figure 5. Running the Security Query

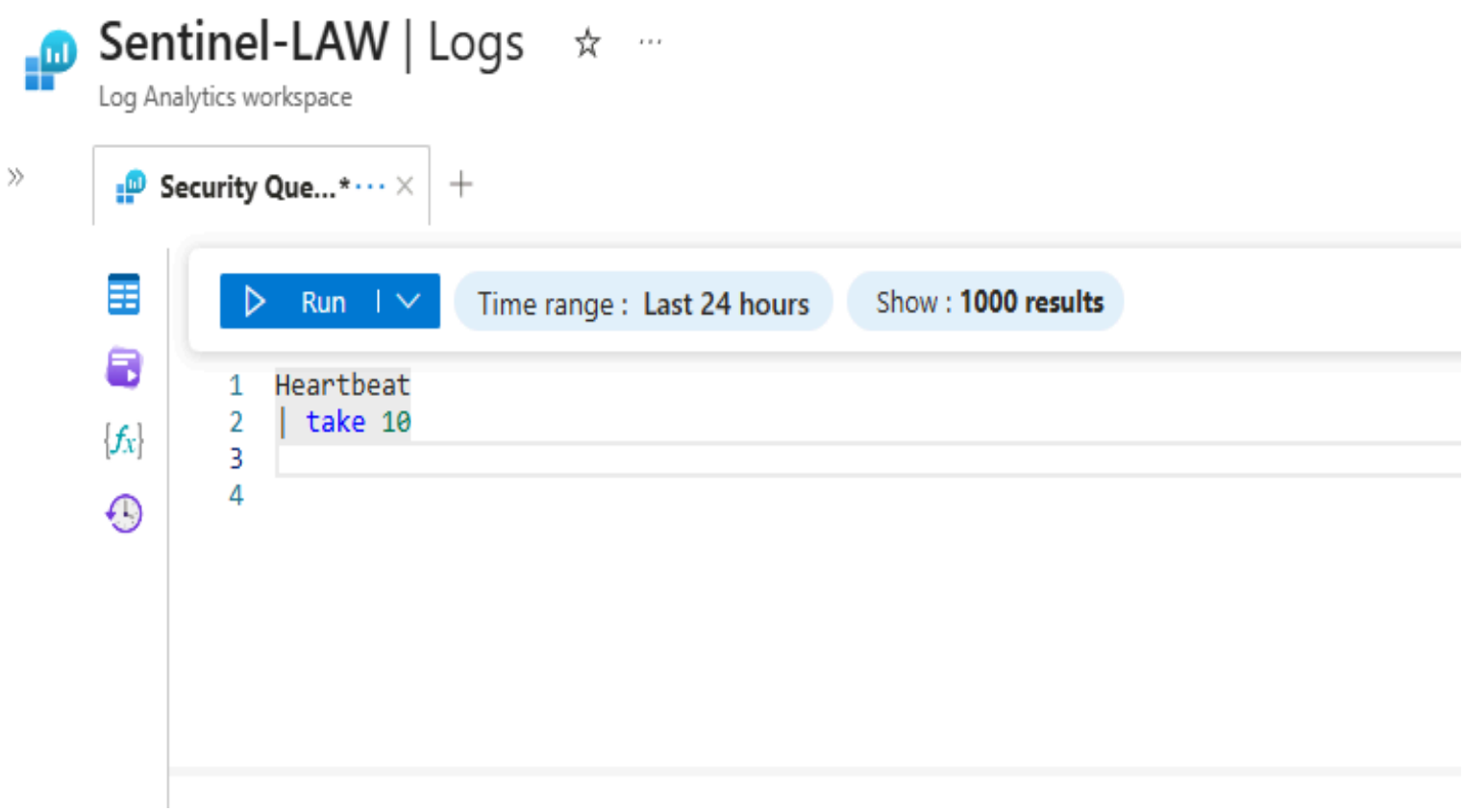


Figure 6. Logs After Running the Query (Column and Chart Format)

Column Format

Results Chart

TimeGenerated [UTC]	SourceComputerId	ComputerIP	Computer	Category	OSType	OSName	OSMajorVersion	OSMinorVersion	V
19/02/2026, 5:03:33.257 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:59:33.217 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:58:33.208 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:57:33.193 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:56:33.182 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:55:33.168 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:54:33.159 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:53:33.168 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:52:33.119 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1
19/02/2026, 4:51:33.109 am	9cd22428-e943-4597-bac0-8eb446df7870	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows	Microsoft Windows Server 2022 Datacenter	10	0	1

Results Chart

SCAgentChannel	RemoteIPLongitude	RemoteIPLatitude	RemoteIPCountry	SubscriptionId	ResourceGroup	ResourceProvider	Resource
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin
Direct	151.21	-33.87	Australia	3036a6d5-0bfd-427a-b7fa-ee4...	sentinel-lab-rg	Microsoft.Compute	myVMDhafin

Chart Format (Figure 6)

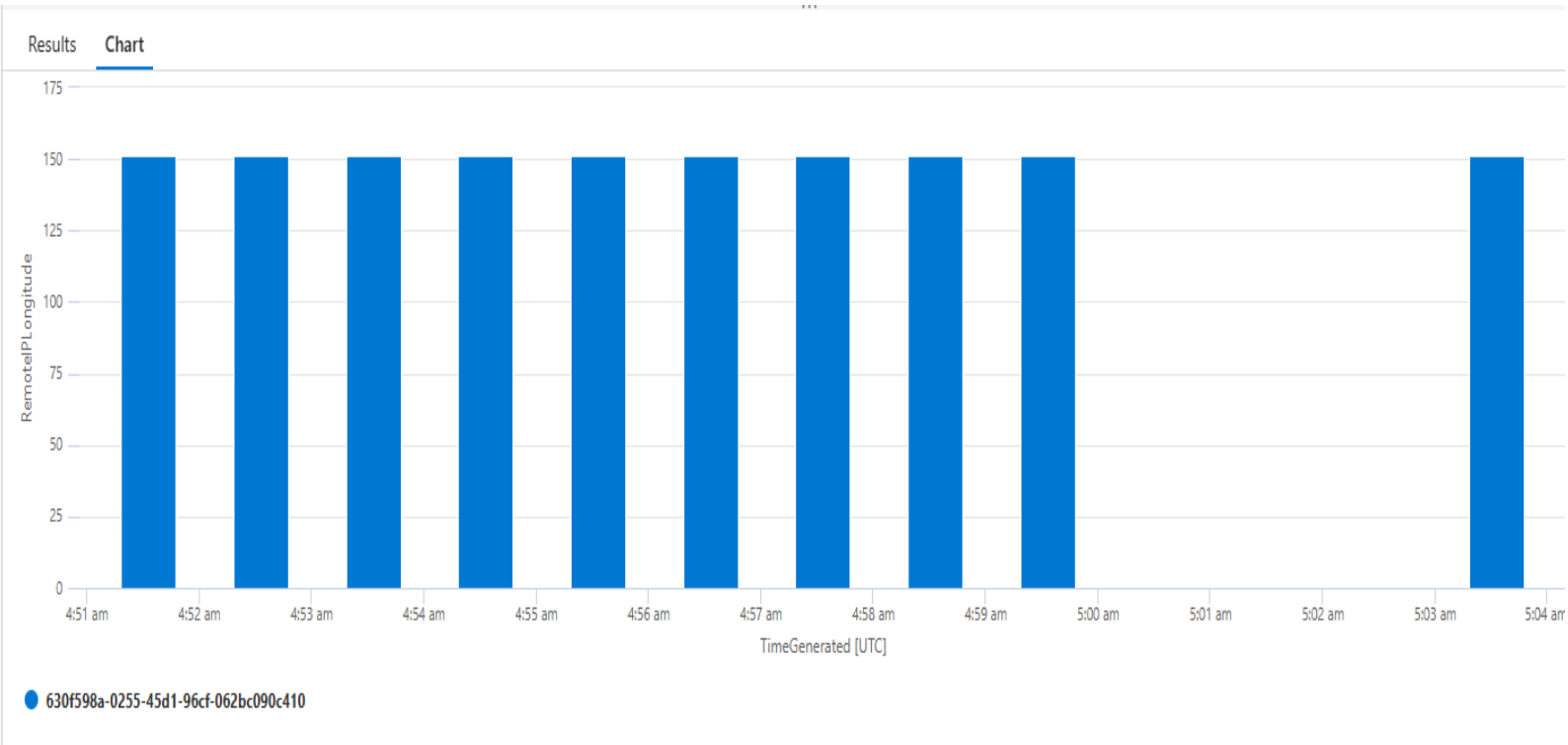
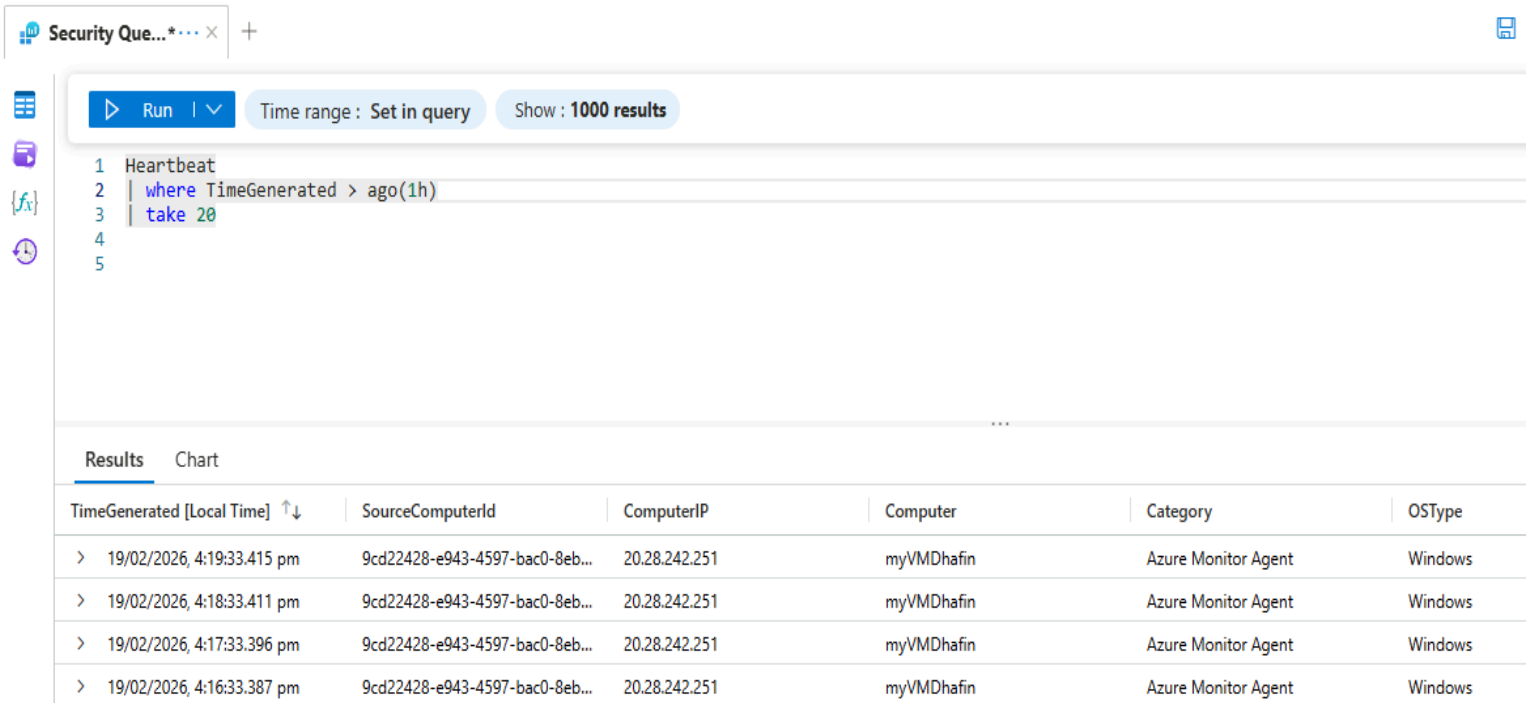


Figure 7. Confirming Security Event Collection and Force Events. (Signing in and Out on Windows VM to Obtain Logon Events) = Heartbeats Logs Received



TimeGenerated [Local Time]	SourceComputerId	ComputerIP	Computer	Category	OSType
19/02/2026, 4:19:33.415 pm	9cd22428-e943-4597-bac0-8eb...	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows
19/02/2026, 4:18:33.411 pm	9cd22428-e943-4597-bac0-8eb...	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows
19/02/2026, 4:17:33.396 pm	9cd22428-e943-4597-bac0-8eb...	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows
19/02/2026, 4:16:33.387 pm	9cd22428-e943-4597-bac0-8eb...	20.28.242.251	myVMDhafin	Azure Monitor Agent	Windows

Phase 6: Enabling Microsoft Sentinel

The Log Analytics Workspace was converted into a cloud-native SIEM with improved threat detection and incident creation capabilities by enabling Microsoft Sentinel.

Figure 1. Adding Microsoft Sentinel to the Sentinel-LAW Workspace

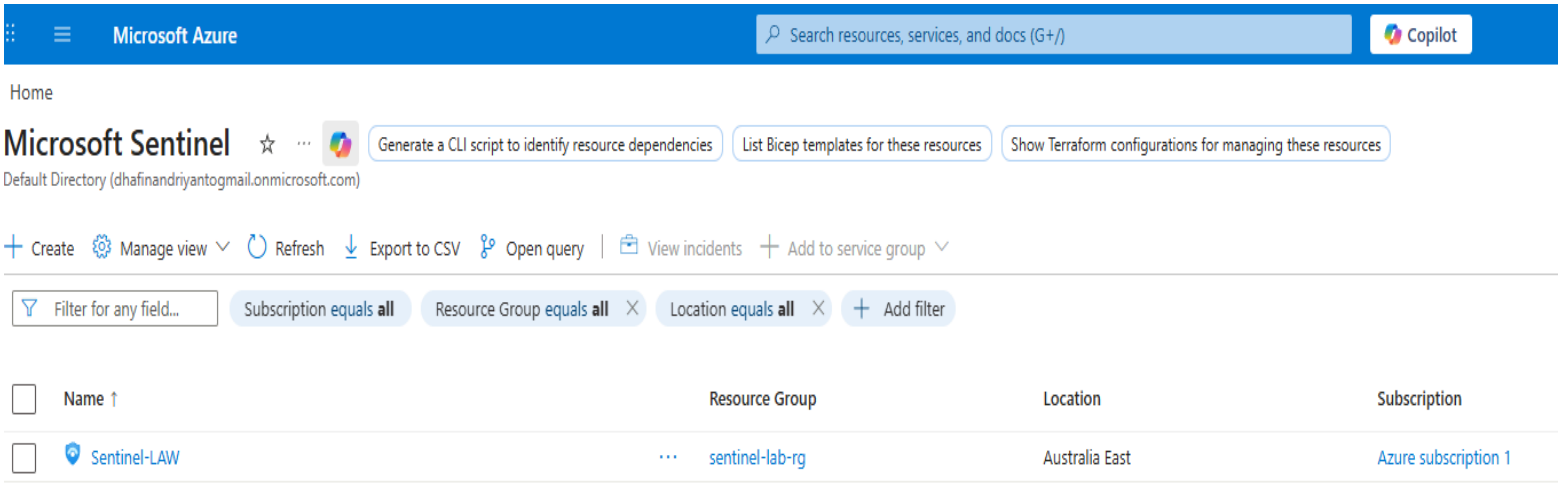


Figure 2. Microsoft Defender Portal (Workspace is Connected)

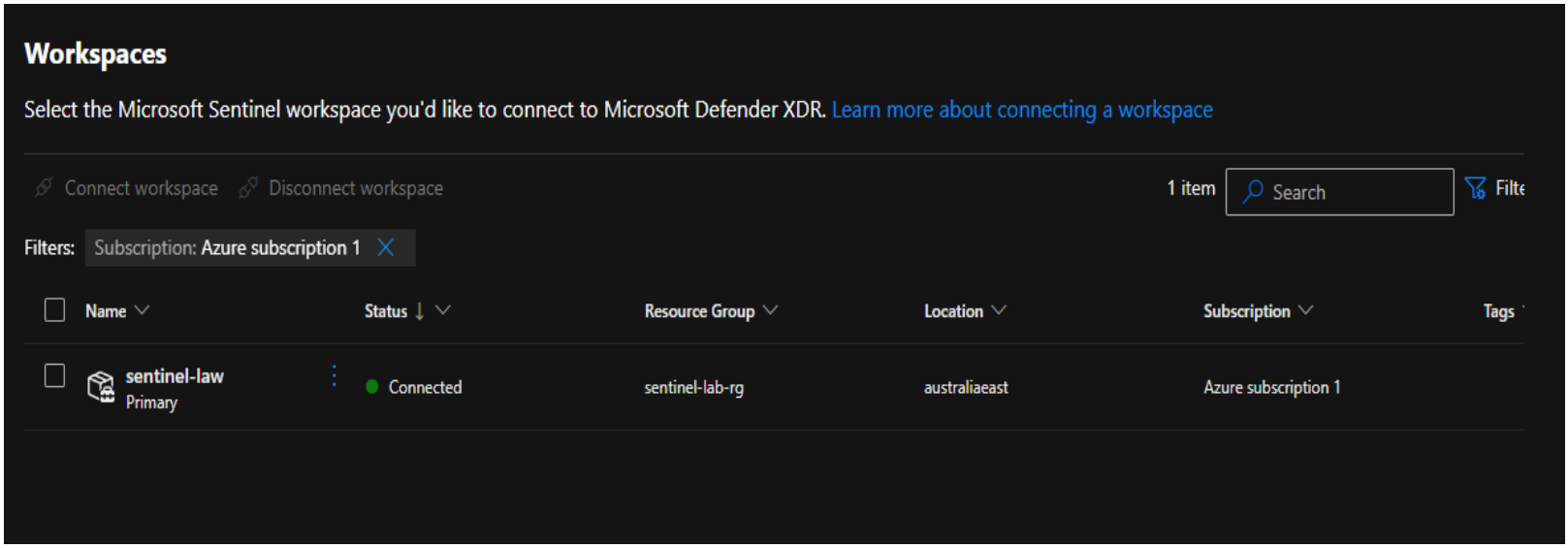


Figure 3. Configuring Windows Security Event Collection (For Security Events) Via Content Hub.

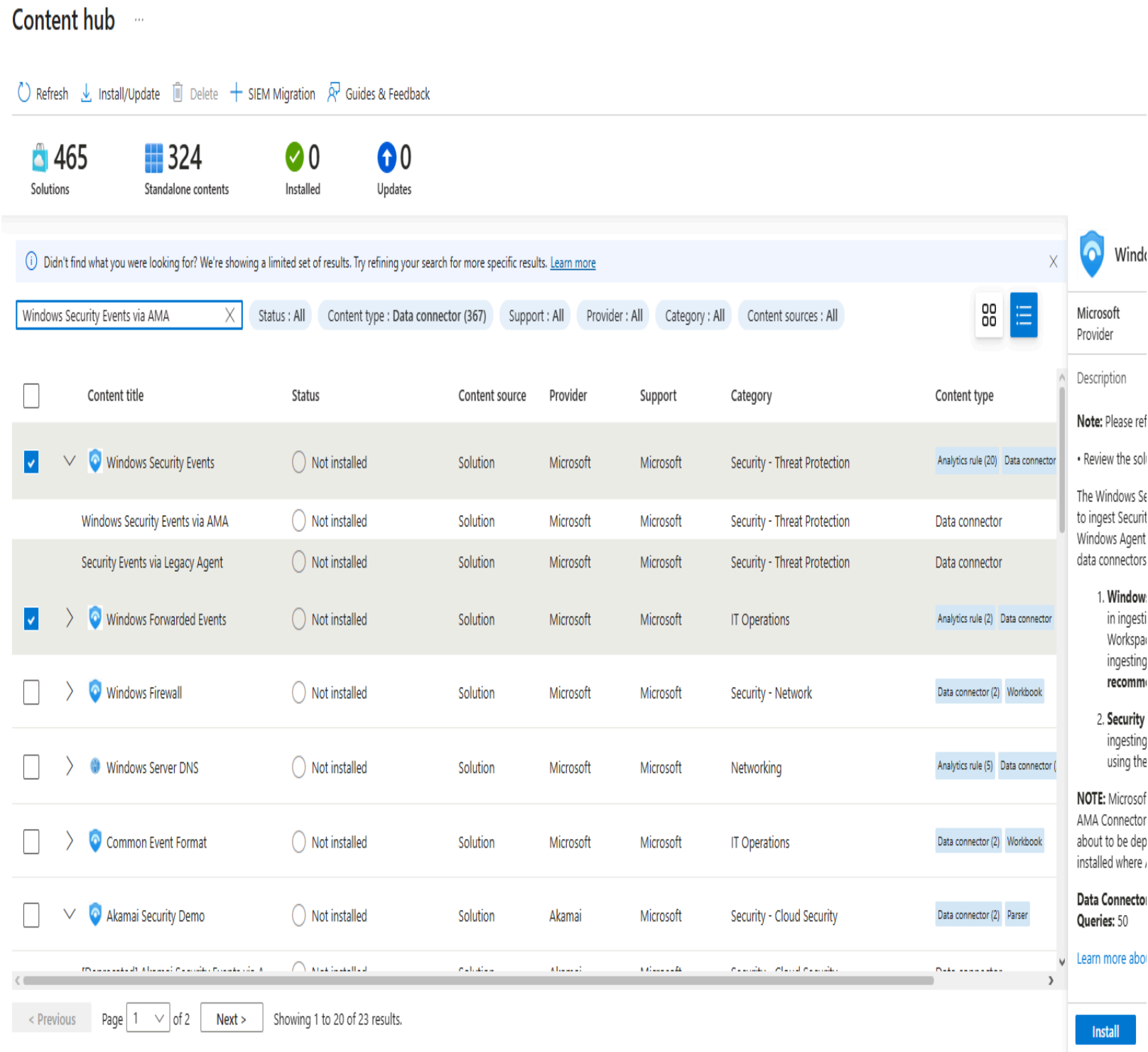


Figure 4. Windows Security Events and Installed Successfully

Content title	Status	Content source	Provider	Support	Category	Content type
Windows Security Events	Installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Analytics rule (20) Data connector
Windows Security Events via AMA	Installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Data connector
Security Events via Legacy Agent	Installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Data connector

Home

Microsoft Sentinel | Data connectors

Selected workspace: 'sentinel-law'

Search

General
- Overview
- Logs
- Guides
- Search

Threat management

Content management

Configuration
- Workspace manager (Preview)
- Data connectors
- Analytics
- Summary rules
- Watchlist
- Automation
- Settings

Refresh Guides & Feedback

Data Connector with "content source = gallery content" have been removed. All the removed content and more is available in content hub. Click here to reinstate in use "content source = gallery content" templ

Device specific AMA connectors have been deprecated. Learn more >

Starting June 2, 2025, the Codeless Connector Platform (CCP) will be renamed to the Codeless Connector Framework (CCF).

9 Connectors 7 Connected More content at Content Hub

Search by name or provider Providers : Microsoft Data Types : All Status : All

Microsoft

Microsoft Defender for Cloud Apps
Microsoft

Microsoft Defender for Endpoint
Microsoft

Microsoft Defender for Identity
Microsoft

Microsoft Defender for Office 365 (Preview)
Microsoft

Microsoft Defender XDR
Microsoft

Microsoft Entra ID Protection
Microsoft

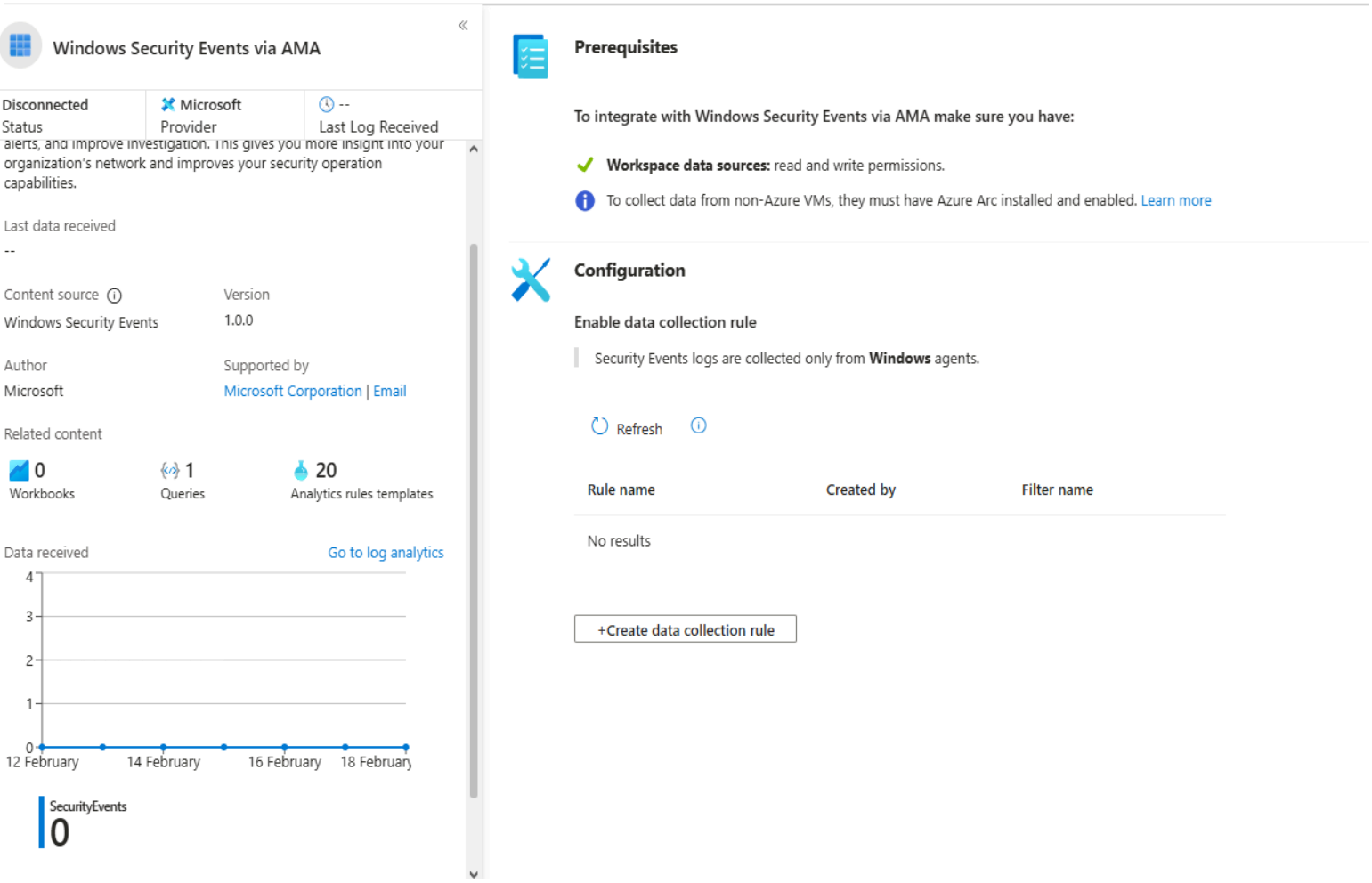
Security Events via Legacy Agent
Microsoft

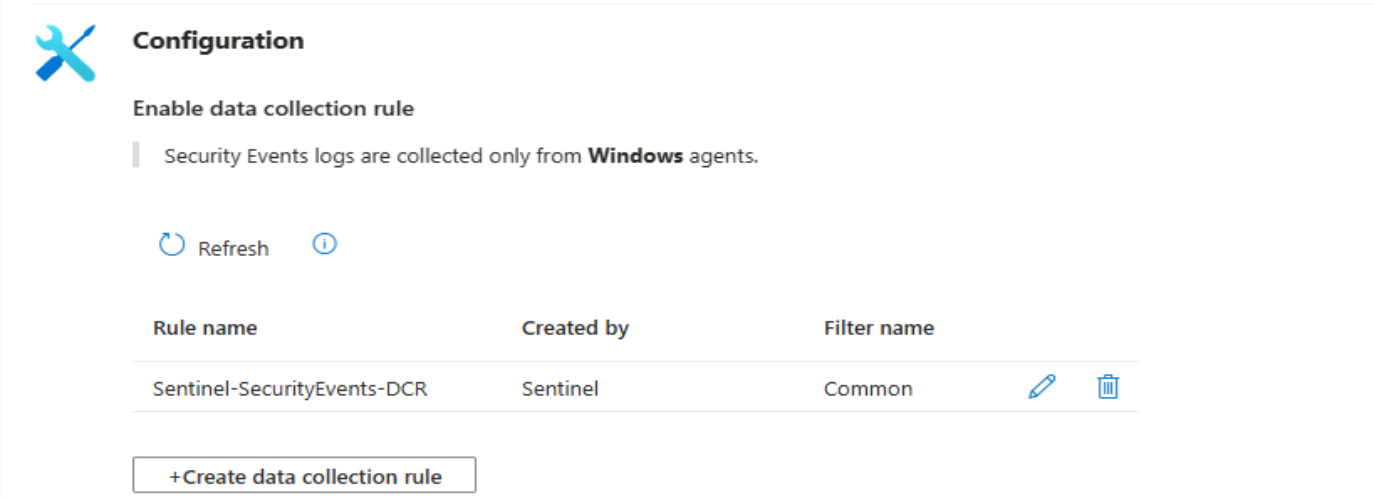
Windows Security Events via AMA
Microsoft

Phase 7: Simulating a Brute-Force Attack

To replicate brute-force behaviour and confirm detection visibility within the SIEM, a number of unsuccessful login attempts were generated against the virtual machine.

1. Creating the Data Collection Rule for the Windows Security Events





2. Windows Security Events (Log) Ingestion via Windows AMA

New Query 1* ··· × +

Save Share ··· Queries h

Run | Time range : Set in query | Show : 1000 results | KQL mode

```
SecurityEvent
| where TimeGenerated > ago(30m)
| take 20
```

Results Chart

TimeGenerated [Local Time]	Account	AccountType	Computer	EventSourceName	Channel
19/02/2026, 7:39:00.456 pm	WORKGROUP\myVMDhafin$	Machine	myVMDhafin	Microsoft-Windows-Security-A...	Security

3. Query to Display Actual Logon Activity (Success: EventID 4624)

Run | Time range : Last 24 hours | Show : 1000 results | KQL mode

```
SecurityEvent
| where EventID in (4624, 4625)
| order by TimeGenerated desc
```

Results Chart

TimeGenerated [Local Time]	Account	AccountType	Computer	EventSourceName	Channel
19/02/2026, 7:44:36.481 pm	NT AUTHORITY\SYSTEM	Machine	myVMDhafin	Microsoft-Windows-Security-A...	Security

4. Same Query Used to Display Logon Activity (Fail: EventID 4625)

Run | Time range : Last 24 hours | Show : 1000 results | KQL mod

```
SecurityEvent
| where EventID in (4624, 4625)
| order by TimeGenerated desc
```

Results | Chart

TimeGenerated [Local Time]	Account	AccountType	Computer	EventSourceName	Channel
19/02/2026, 11:55:55.522 pm	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
19/02/2026, 11:55:52.771 pm	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
19/02/2026, 11:55:48.267 pm	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
19/02/2026, 11:55:44.809 pm	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security

EventID	Activity	AuthenticationPackageName	ElevatedToken	ImpersonationLevel	IpAd
4625	4625 - An account failed to log on.	NTLM			101.1
4625	4625 - An account failed to log on.	NTLM			101.1
4625	4625 - An account failed to log on.	NTLM			101.1
4625	4625 - An account failed to log on.	NTLM			101.1

Phase 8: Creating an Analytics Rule

Using KQL logic, a scheduled analytics rule was developed to identify recurring unsuccessful login attempts and automatically generate a security incident in Microsoft Sentinel.

Figure 1. Analytics Page:

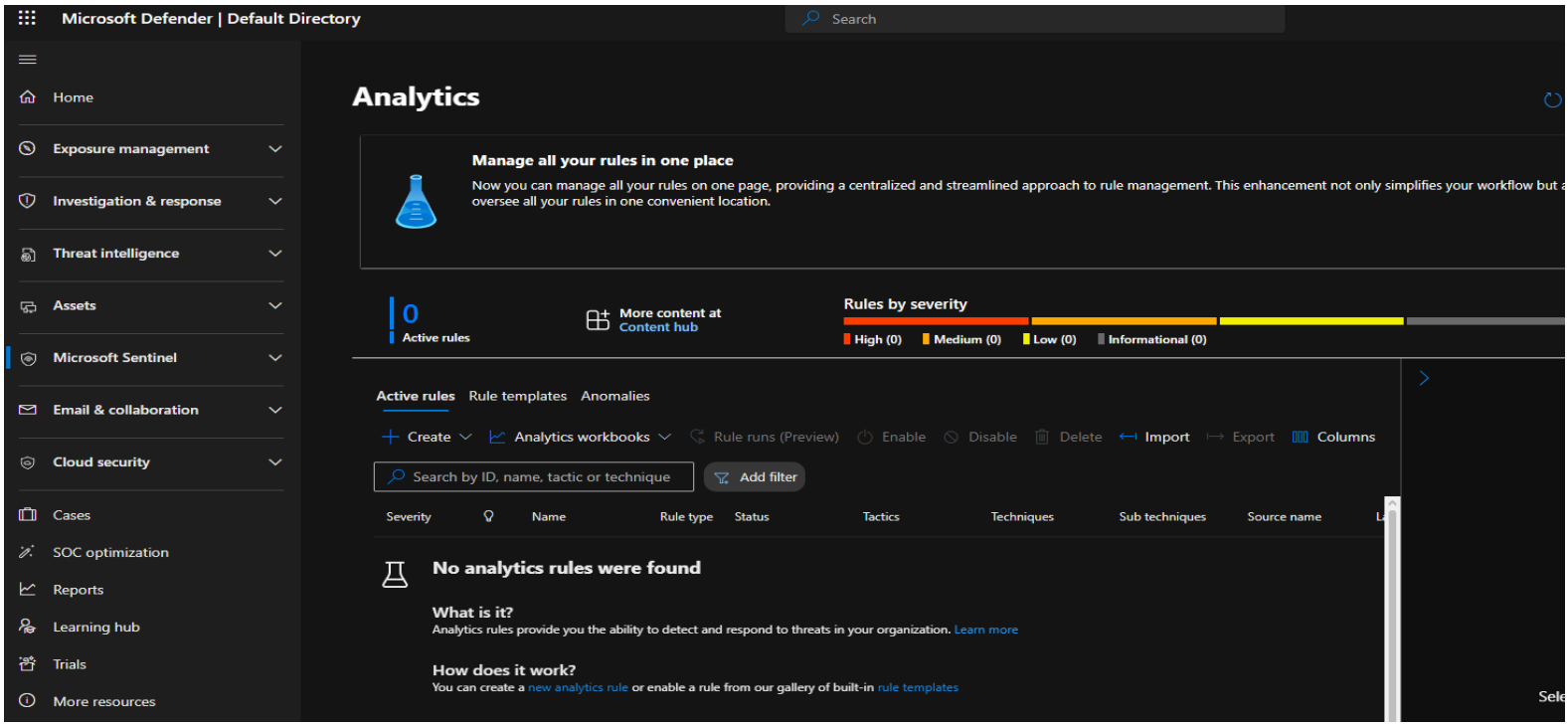
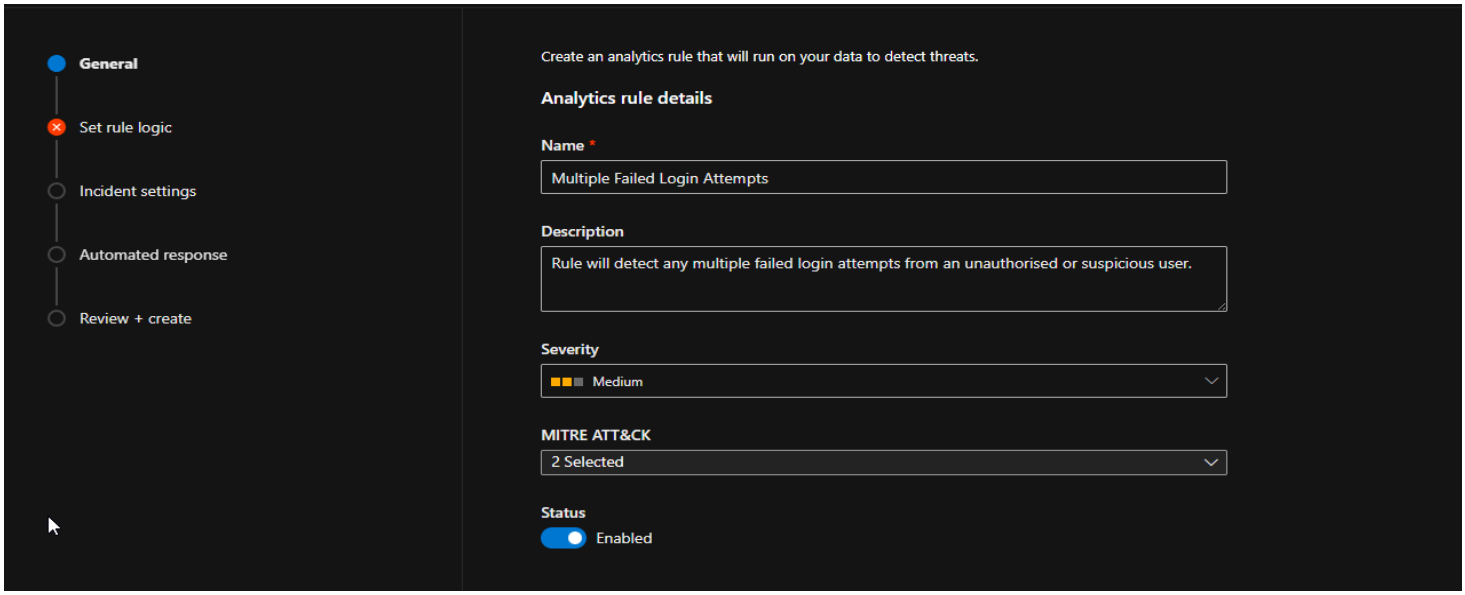
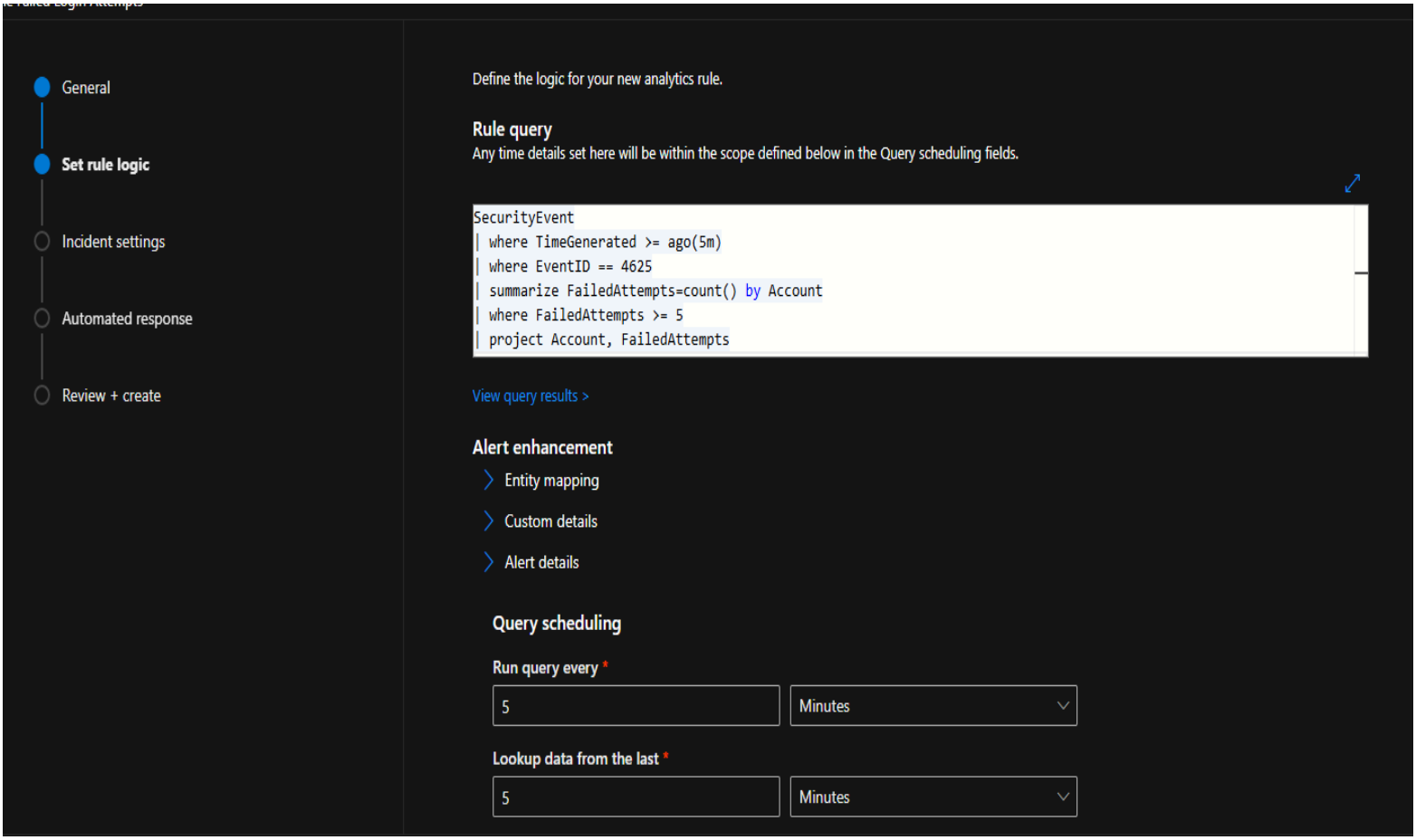


Figure 2. Creating a New Scheduled Rule (General, Set Logic Rule, Incident Settings, and Automated Response)

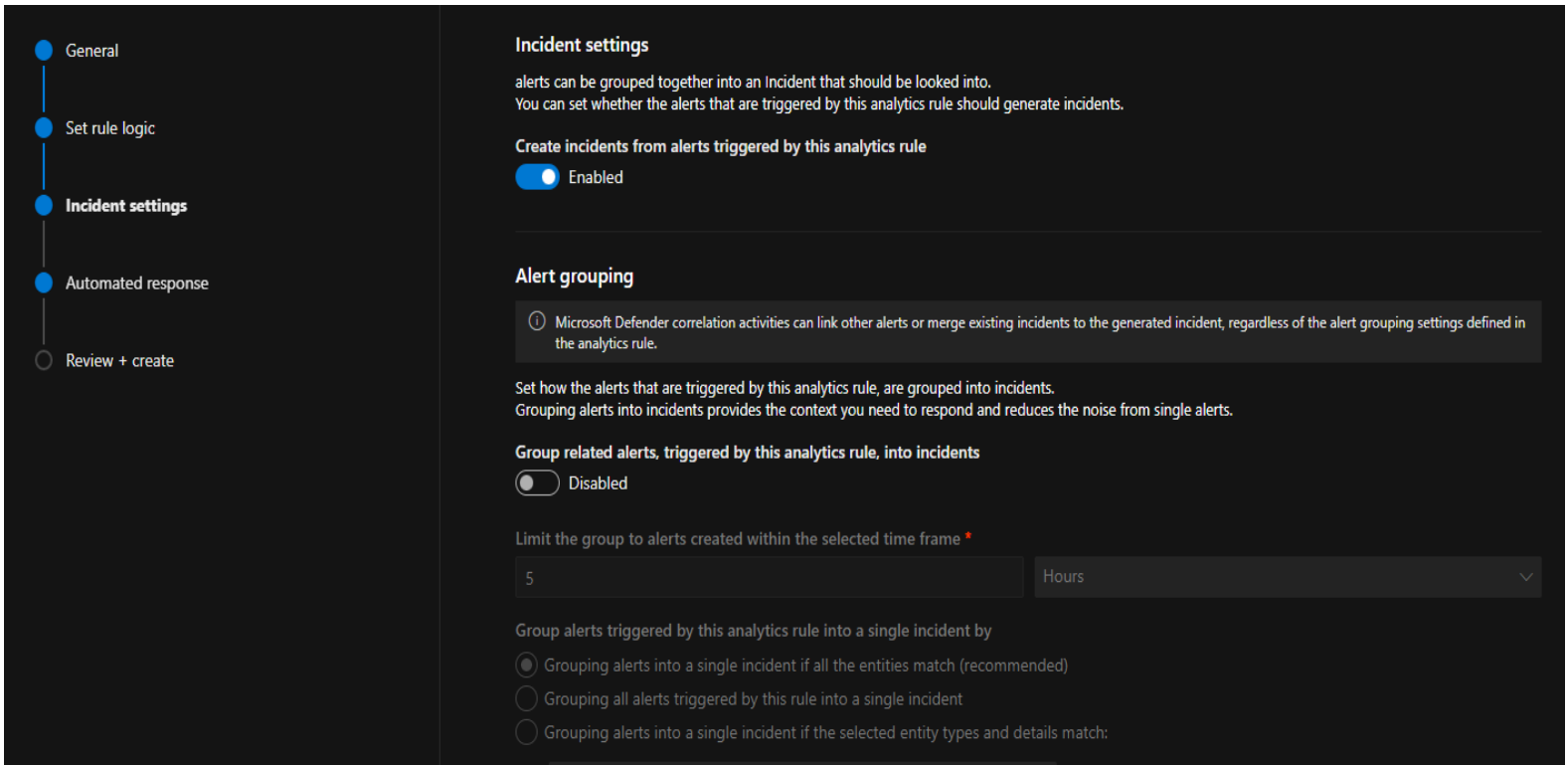
- **General Screenshot**



- **Set Logic Rule Screenshot**



- **Incident Settings Screenshot**



- **Automated Response Screenshot**

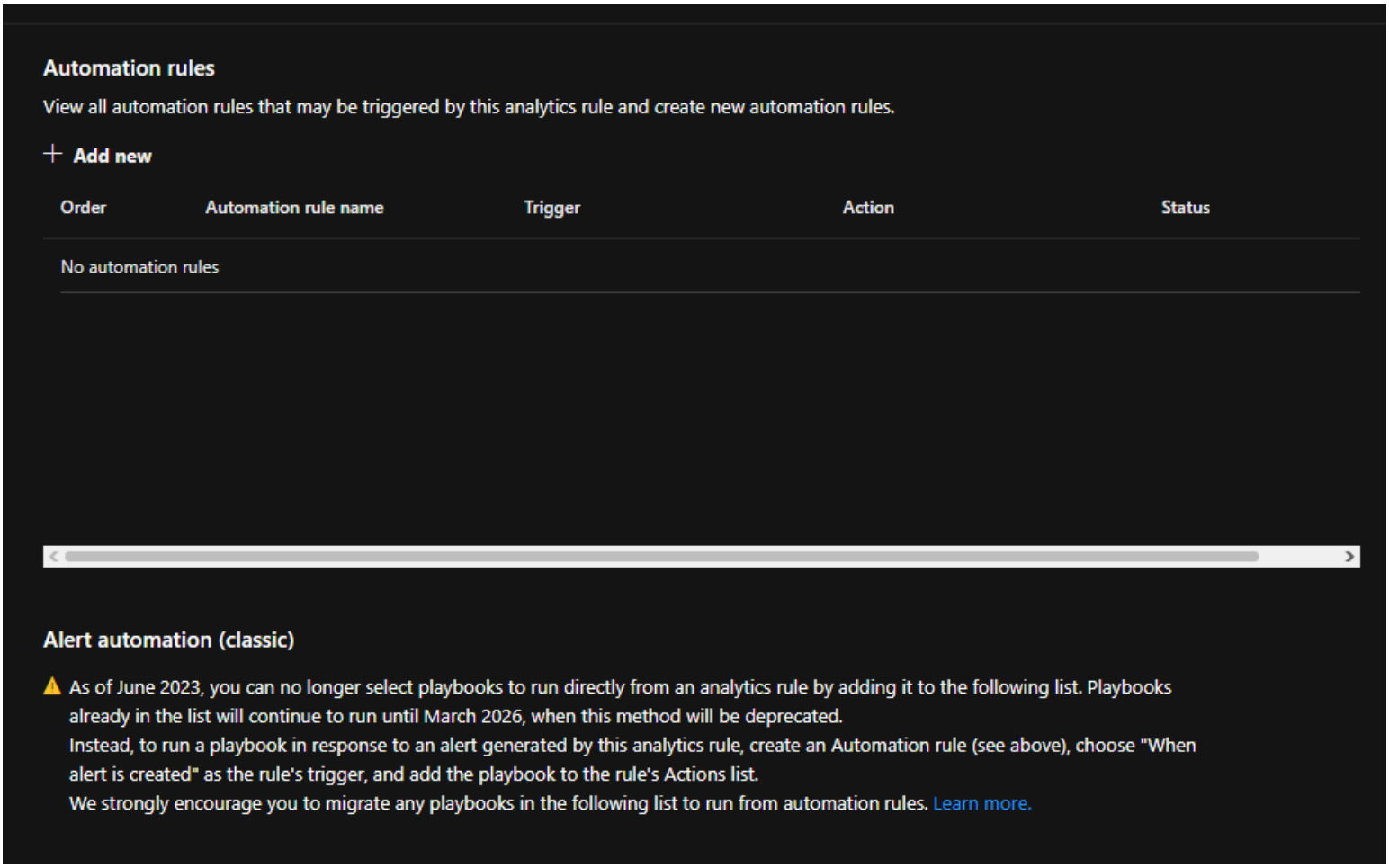
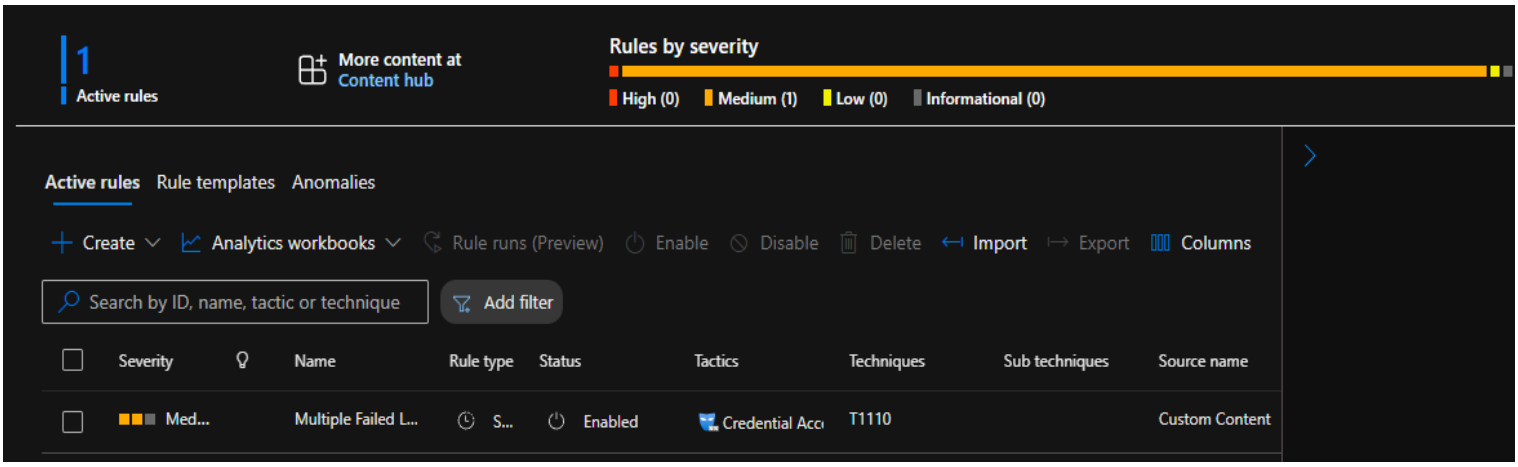


Figure 3. Deployment of Analytics Rule



Phase 9: Incident Investigation

Figure 1. Incident Alert on Incident Page (Multiple Failed Login Attempts)

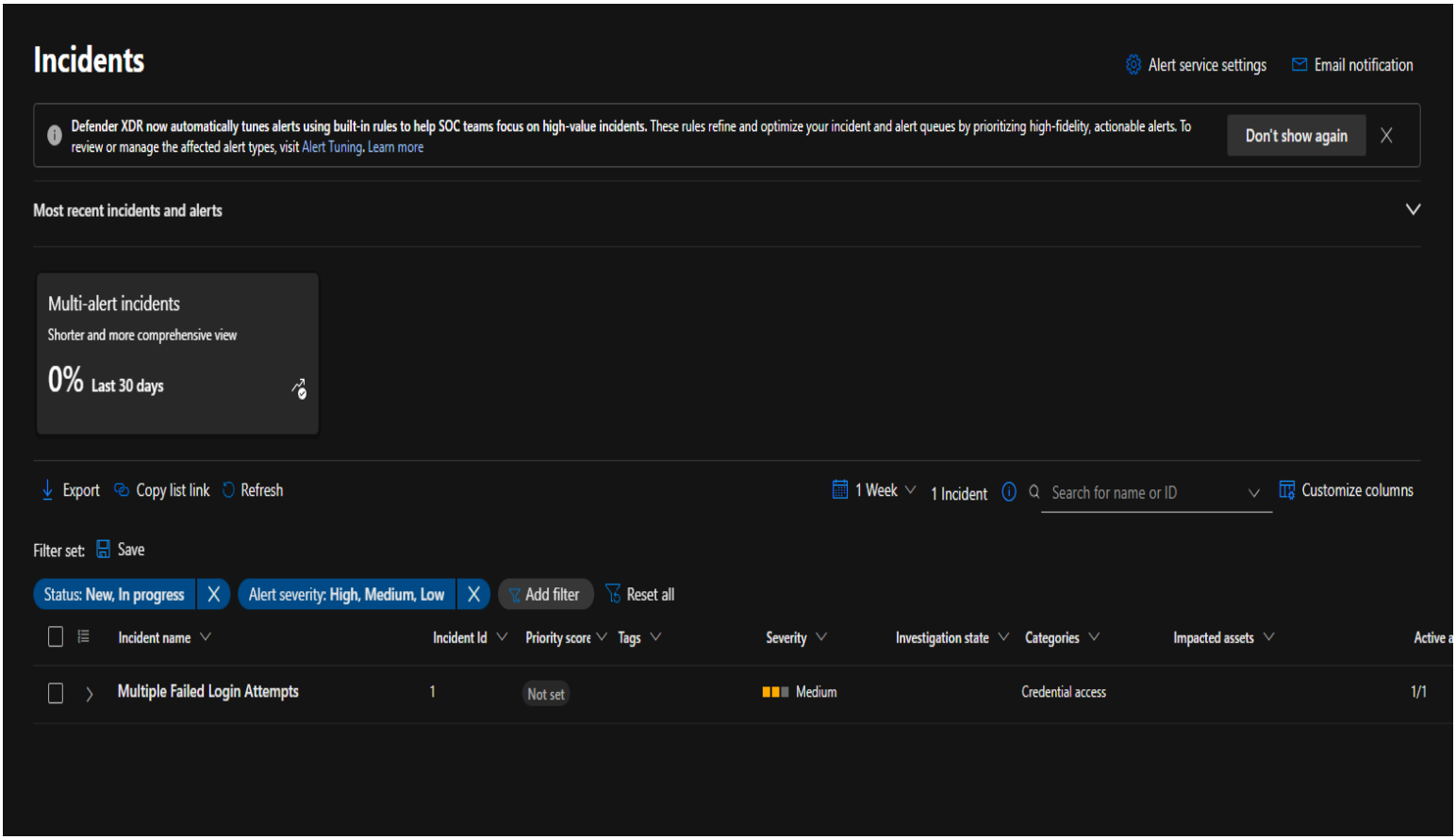


Figure 2. Alert Details of Multiple Failed Login Attempts

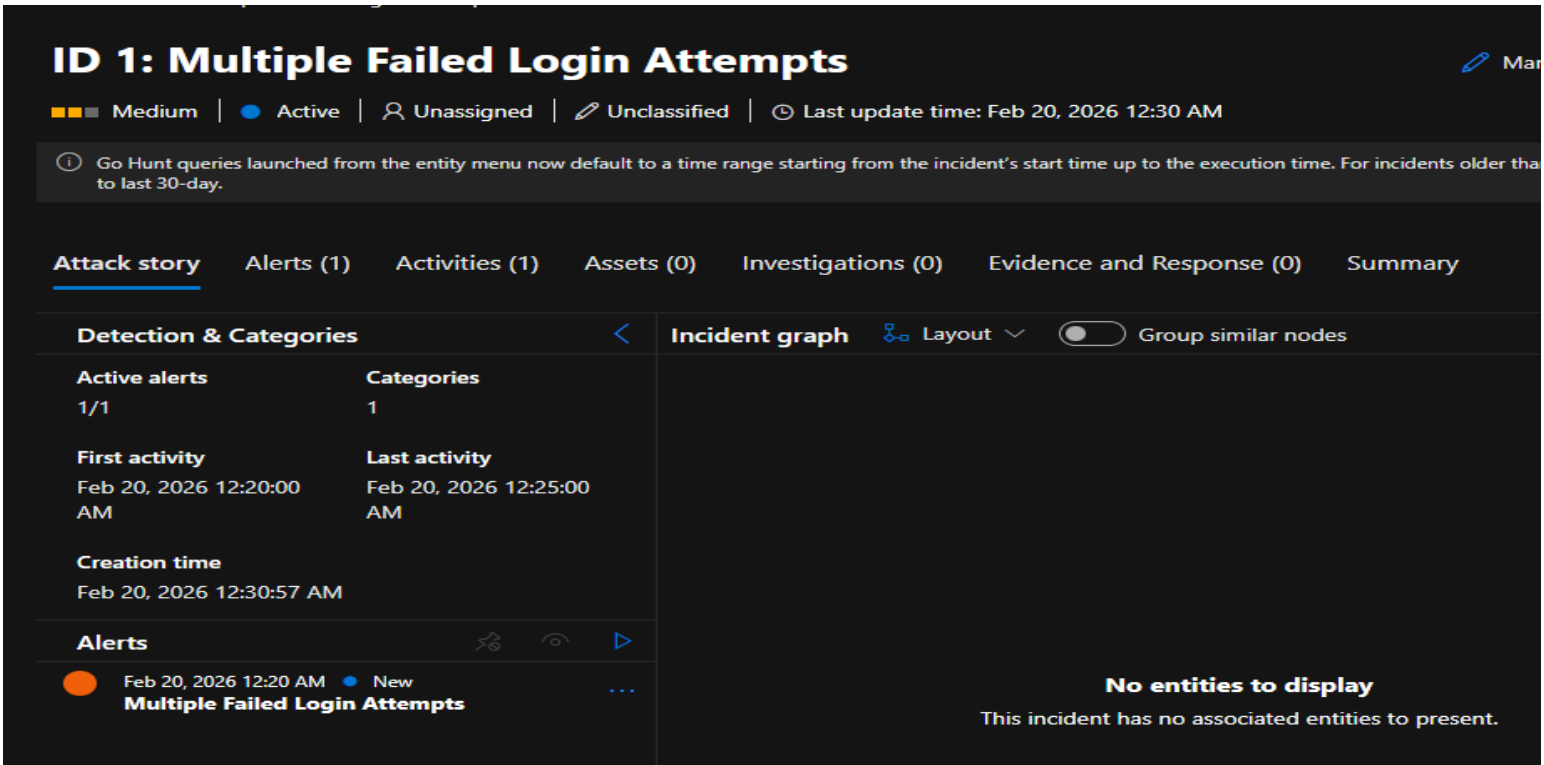


Figure 3. (Screenshot Taken From Phase 3, Figure 2). To Simulate Brute-Force Attacks. Incorrect Credentials are Entered on this Login Page For the Windows VM

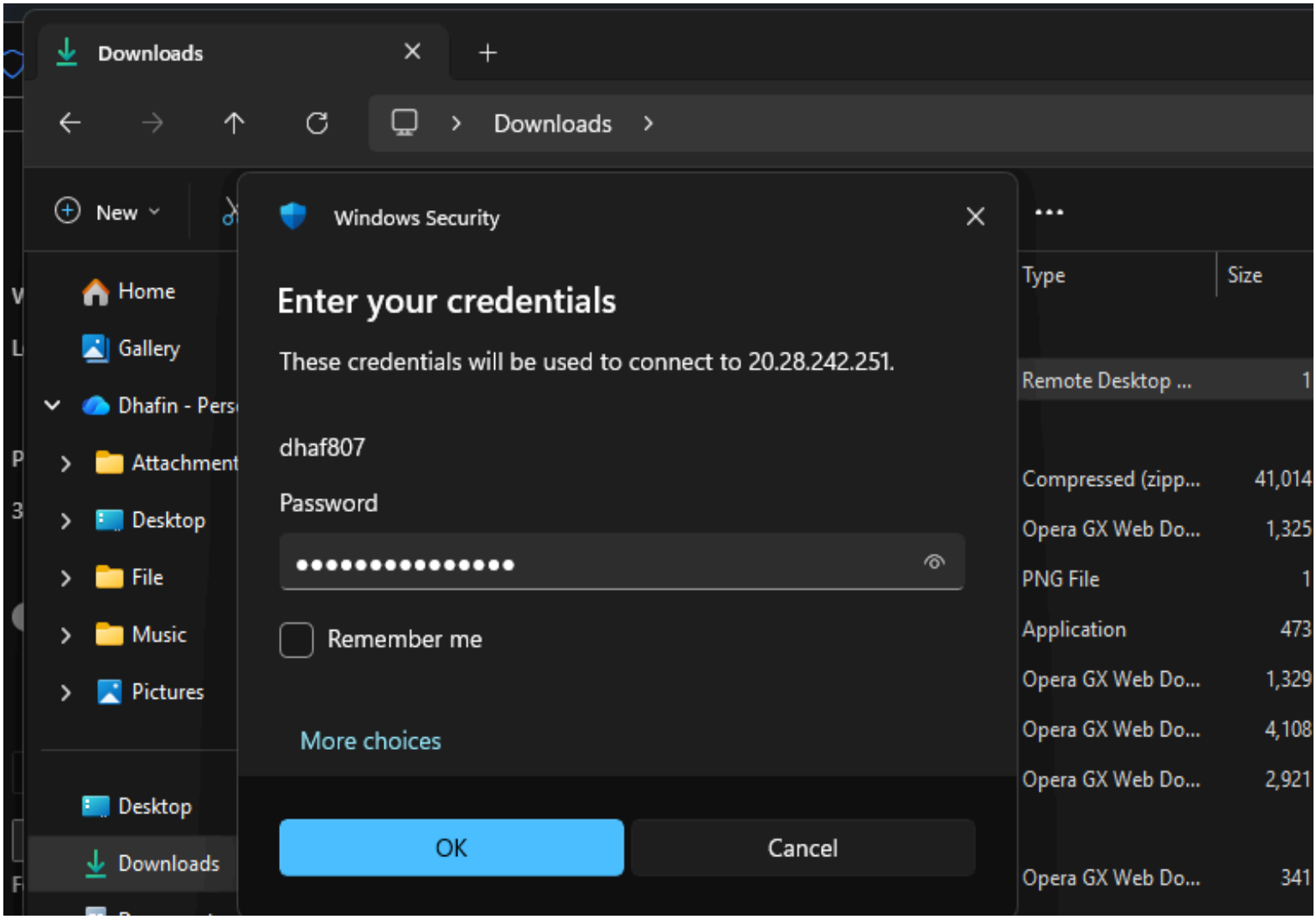


Figure 4. Raw Log Evidence of Login Failure into Windows Virtual Machine (4625)

New Query 1* ··· × +

Save Share ···

Run | Time range : Set in query | Show : 1000 results | KQL

```
SecurityEvent
| where EventID == 4625
| where TimeGenerated >= ago(30m)
```

Results Chart

TimeGenerated [Local Time]	Account	AccountType	Computer	EventSourceName	Channel
20/02/2026, 12:25:35.463 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:25:34.194 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:25:31.011 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:25:29.664 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:25:27.674 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:25:25.513 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:24:22.568 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:24:20.493 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:24:17.739 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:24:12.571 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security
20/02/2026, 12:24:09.356 am	MicrosoftAccount\dhaf807	User	myVMDhafin	Microsoft-Windows-Security-A...	Security

Results Chart

EventID	Activity	AuthenticationPackageName	FailureReason
4625	4625 - An account failed to log on.	NTLM	%%2307
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313
4625	4625 - An account failed to log on.	NTLM	%%2313

Detection Logic Design

Windows Security Event ID 4625 (failed logon attempts) was configured to be monitored by the scheduled analytics rule. When several unsuccessful authentication attempts occurred within a specified time frame, the detection logic was designed to activate, indicating behaviour representative of brute-force activity.

The rule was configured with:

- A specified time frame for lookback
- A minimum number of failed login attempts
- Enabled automatic incident creation

This setup converts authentication telemetry into actionable security incidents, simulating real-world SOC alerting operations.

Considerations for Security Hardening

Production environments need preventative security controls to reduce brute-force vulnerability, even though this lab focused on detection capabilities.

These may include:

- Implementing account lockout procedures
- Multi-factor authentication (MFA) implementation
- Using Network Security Groups (NSGs) to limit RDP access
- Integrating Microsoft Defender for Endpoint
- Applying Conditional Access policies into practice

By lessening the possibility and effect of credential-based attacks, these restrictions complement detection mechanisms.

Lessons Learned

This project reinforced several crucial security concepts.

- The requirement to verify log ingestion prior to creating detection logic
- The importance of understanding Windows Security Event IDs for the purpose of monitor authentication

- How Sentinel, Azure Monitor Agent, and Data Collection Rules collaborate in a cloud logging pipeline
- The effect of accurate configuration on the generation of successful alerts

The lab focused on the connection between effective threat detection and cloud infrastructure visibility.

Conclusion

This project effectively demonstrated how to use a cloud-native SIEM system in Microsoft Azure to identify attempts at brute-force authentication against a virtual Windows computer. Repeated unsuccessful attempts to log in were converted in Windows Security Events and applying scheduled KQL-based analytics rules within Microsoft Sentinel.

After threshold-based authentication failed, an incident was successfully generated, validating the entire detection pipeline, from telemetry collection to alert generating and investigation workflow. This confirms that the environment was configured appropriately and that it could detect credential-based attack patterns in an infrastructure hosted in the cloud.

In summary, this lab highlights the importance of automated alerting, detection engineering, and centralised logging in cloud security operations. It demonstrates how authentication telemetry can be operationalized into valuable security insights in an advanced SIEM platform.